

Summarized chain of thought (Quasipolynomial bounds for arithmetic progressions)

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The two tasks seek progressively stronger density bounds for arithmetic progressions. The first develops an argument for the reciprocal-sum problem; the second seeks a quasipolynomial bound for every fixed progression length, using the previous reciprocal-sum bound and reorganizing the precision costs of successive density increments.

Part I: The reciprocal-sum problem

Original prompt (excerpts)

Problem

Let $N_0 = \{0, 1, 2, \dots\}$, and let A be a subset of N_0 such that $\sum_{n \in A, n \geq 1} 1/n$ diverges to infinity.

Prove or disprove that A contains arithmetic progressions of every finite length: for every integer $k \geq 1$, there are integers $a \geq 0$ and $d \geq 1$ such that $a, a+d, \dots, a+(k-1)d$ all belong to A . The requirement $d \geq 1$ makes the progression nontrivial for $k \geq 2$.

A disproof must provide a set with divergent reciprocal sum and some fixed length $k \geq 3$ for which no such progression exists.

1 Seeking structure without losing the density gain

The assistant begins by asking whether a divergent reciprocal sum forces arithmetic progressions of every finite length. It uses Bloom and Sisask's three-term reciprocal-sum result [3] as a baseline, alongside Green and Tao's polylogarithmic bound for $r_4(N)$ [17] and Gowers's general quantitative bound [13], then seeks stronger bounds for longer progressions.

Let $r_k(N)$ be the largest size of a subset of $\{1, \dots, N\}$ with no nontrivial k -term progression. Convergence of

$$\sum_{j \geq 1} \frac{r_k(2^j)}{2^j}$$

would suffice for the reciprocal-sum conclusion. Qualitative density decay alone does not supply this summability. Reading the Leng–Sah–Sawhney bound [35] directs attention to losses incurred when structured density increments are converted into ordinary progressions. The underlying quasipolynomial inverse theorem [36] offers controlled nilsequence dimension, but the assistant identifies the accumulation of many weak correlations as a further cost. The assistant proposes retaining structured ambient sets through successive increments and borrowing sparse counting and high-moment ideas from Kelley–Meka methods [33].

For sparse counting it studies Filmus, Hatami, Hosseini, and Kelman's grid norms and binary-system counting lemma [12], seeking a higher-order analogue. It also consults the relative sifting argument for corners of Jaber, Liu, Lovett, Ostuni, and Sawhney [28], particularly the need to avoid losses from the density of a sparse ambient set. The proposed extension encounters several obstacles. The positivity

mechanism useful for three-term progressions does not directly extend to longer rooted counts. A tensor-product sketch questions whether fixed uniformity norms can control sparse normalized functions, but the assistant does not complete the sketch. The assistant therefore turns toward relative correlations, higher moments, and compression of many dependent polynomial phases.

Bloom and Sisask’s exposition [4] supplies the unbalancing, sifting, and almost-periodicity framework, including the Schoen–Sisask almost-periodicity input [49]. The assistant tries to combine the linearization and symmetry steps familiar from the U^3 inverse theorem [18] with Croot–Sisask smoothing [9], while preserving relative correlation on a selected set of shifts. For higher orders it considers quantitative inverse theory, including Manners’s bounds [41], as a possible ingredient without completing that extension. The assistant proposes aligning several frequency functions simultaneously, instead of accumulating losses through sequential alignment:

VERBATIM EXCERPT

No iterative losses if align all frequencies at once. Great. Need ensure S_h can be recovered from frequencies with large correlations $\geq \exp(-\text{poly})$.

This would organize correlations of pair products $f(x)f(x+h)$, where $f = 1_A/\alpha$ and $\alpha = |A|/N$. The assistant still needs to preserve relative correlation and complete the counting argument.

The assistant then identifies an obstruction: upper density bounds on structured pieces permit rare empty pieces. Centered products can concentrate on those holes. Adding a constant background briefly appears to furnish a pseudorandom majorant, but the resulting error is too large relative to the diluted density. Adaptive filling introduces further feasibility conditions that the assistant has not proved:

VERBATIM EXCERPT

However majorant on **all** rare atoms likely unrealistic due to impossible upper and lower simultaneously for constructions with systematic holes.

It returns to simultaneous frequency alignment while continuing to seek a counting lemma.

2 Sparse counting, product obstructions, and an entropy route

With $\mu = 1_A/\alpha$, $\mathbb{E}\mu = 1$, and $M = \alpha^{-1}$, the central question was whether upper bounds on structured concentration could force positive progression counts with affordable dependence on M . The assistant considered Green–Tao transference [20], inspected Zhao’s arithmetic transference proof [53], and studied Conlon–Fox–Zhao densification [7]; it found that the latter relied on two-sided linear-forms estimates. The sparse graph counting lemma of Filmus, Hatami, Hosseini, and Kelman [12] supplied the model for asymmetric grid moments and positive-semidefinite unbalancing.

Upper bounds alone repeatedly encountered sparse holes. Large centered grid moments could reflect missing mass rather than upward concentration, while weighted Cauchy–Schwarz retained factors that obstructed induction. The assistant considered allowing small power losses in M if localization costs accumulated additively, but obtained no suitable counting theorem. It also considered Croot–Sisask smoothing [9], finding its required accuracy too costly in the proposed application.

Product amplification offered another route. A sufficiently strong recurrence for extremal progression-free densities might yield summability across exponential scales. Fiber bounds did not multiply automatically, however, and constant-norm spheres obstructed the hoped-for strong decay over fixed digit alphabets. The assistant compared Green–Tao four-term bounds over finite fields and integers [19, 17] with the Leng–Sah–Sawhney progression bound and quasipolynomial inverse theorem [35, 36]; attempted

lifting and iteration still lost too much quantitatively.

Returning to hypergraph links, the assistant proposed high-moment control of common-neighbor degrees:

VERBATIM EXCERPT

Excellent, products degree fluctuations handled with polynomial precision! Then for fixed u , H' path common-neighbor also concentrated for y, z even weighted. Can we use to control signed g' vs path distribution with required precision c^p ?

The assistant noticed that signed expansions could have absolute mass of order 2^p , overwhelming polynomial accuracy. The assistant explored using the quasipolynomial corners bound of Jaber, Liu, Lovett, Ostuni, and Sawhney [28] on parameters of partial progressions, but the resulting corners did not automatically assemble longer progressions, while positivity of the four-term box count did not control its exceptional diagonal.

The investigation then turned toward entropy at scale $p = \log M$, drawing on the Kelley–Meka use of high moments [33]:

VERBATIM EXCERPT

So can leverage entropy instead of L^2 by nonnegativity and frequency dissociative (Chang). Kelley–Meka essentially accounts for many tiny biases via L_p .

The assistant next sought a Fourier restriction estimate $\|\widehat{\mu g}\|_q = O(1)$ for bounded g and fixed $q > 2$, under suitable local upper bounds. Independent-frequency estimates suggested a starting point, but extracting structure from high moments lost too much mass. The final calculation tried to express sampled-character moments positively so local bounds could apply. The assistant left the calculation unfinished, without deriving either the restriction lemma or the sparse counting principle.

3 Endpoint restriction and the limits of sparse counting

With normalized density $\mu = 1_A/\alpha$ and logarithmic sparsity $p = \log(1/\alpha)$, the aim is to exploit upper density bounds on small structured factors without repeatedly paying powers of $1/\alpha$.

The assistant first tries Fourier restriction for four-term progressions. The assistant considers the Ben-Sasson–Ron–Zewi approximate-duality approach [2] to extract structured frequency subsets, but finds the contemplated iteration loses too much quantitatively. Truncating rare coherent peaks works for independent frequency clusters, but extending that argument to dependent clusters remains unresolved. The assistant uses Kelley–Meka convolution estimates [33] and high convolution moments to propose an endpoint Fourier ℓ^3 estimate:

VERBATIM EXCERPT

Aha! Exactly weak ℓ^3 from convolution and cap. Need strictly better (<3) for interpolation.

The calculation subsequently reaches a full endpoint bound under its convolution hypotheses, but fails to cross the exponent-three threshold needed by the proposed counting argument. Longer progressions introduce further higher-order obstructions.

The assistant then asks whether upper counts alone can force lower progression counts. Complete box norms offer a possible mechanism, whereas negative cubic contributions obstruct a straightforward extension to progression systems. A separate reduction to colored additive matchings encounters a proposed linear-algebraic barrier: for t blocks with m parameters each, generic diagonal scaling forces the

relation rank to satisfy $R \geq mt/2$. The assistant finds that this defeats the specified closure construction's intended slice-rank saving.

It also considers discretizing Keleti's full-dimensional progression-avoiding construction [31], but does not obtain the logarithmic-density lower bound that this counterexample route would need.

Returning to sparse counting, the assistant revisits the quasipolynomial inverse theorem of Leng, Sah, and Sawhney and their density-increment argument [36, 35]. It calculates how repeated progression restrictions can exhaust the available length, and recalls Green and Tao's density-increment approach in *New bounds for Szemerédi's theorem, II* [21] while considering whether stronger increments or localization could reduce this loss.

It also reexamines Conlon, Fox, and Zhao's densification argument [7]: the required two-sided linear-forms information would already include the lower counts sought here, so the assistant does not obtain them from upper bounds alone. Returning to Bloom and Sisask's exposition of Kelley–Meka [4], it studies whether sifting and almost-periodicity can instead supply a suitable positive convolution comparison.

The assistant identifies two linked difficulties in sparse counting. Repeated density increments can consume too much progression length, while greedy removal of dense cosets creates holes that manufacture further apparent increments. The assistant therefore considers a bounded Fourier model $0 \leq g \leq 1 + O(\varepsilon)$ approximating a nonnegative G in Fourier ℓ^3 . In an adaptive partition calculation, it observes:

VERBATIM EXCERPT

| Great! Norm $q \geq 2$ localizes by averaging across disjoint equal subspace partitions.

Allowing different refinement directions in different cells might control complexity along each branch. Yet an increment can occur on an extremely rare child, leaving typical branches almost unchanged and refinement depth uncontrolled. The assistant leaves the proposed dense-model theorem and the needed quantitative counting estimates conjectural, with refinement depth still uncontrolled.

4 Polynomial factors and lower progression counts

Most calculations used a finite-field model: a nonnegative density μ of mean one, bounded by $M = e^p$, and its centered function $g = \mu - 1$. The desired structural hypothesis prohibited large density increases on suitably simple cells.

An initial proposal used Kelley–Meka methods [33] to turn Fourier concentration into an absolute variance increment. The assistant revisited Bloom and Sisask's exposition [4], using its almost-periodicity statement, drawn from Schoen and Sisask [49], to seek a subspace preserving mass on a large-autocorrelation set. Adaptive partitions and polynomial-phase alignment were then intended to keep approximation costs polynomial in p . For polynomial factors, it invoked the polynomial bounds relating partition rank and analytic rank of Janzer and Milićević [29, 44], proposing to replace biased top-degree combinations by lower-degree polynomials. The quantitative inverse estimates needed for phase alignment were less certain: the assistant recalled Gowers and Milićević's U^4 inverse theorem [14] and Milićević's work on approximate quadratic varieties [45], while leaving its proposed general finite-field estimate conditional. It also compared the intended complexity costs with the inverse theorem and density-increment bounds of Leng, Sah, and Sawhney [36, 35]. The assistant also observes that upper bounds could fail to detect missing configurations:

VERBATIM EXCERPT

| Testing upper over subfunctions and anchoring constants can't rule holes.

For progression counting, eliminating a sparse factor incurred a loss of order M^{-1} . It then tries interpolating moments of

$$Z(y_1, \dots, y_q) = \mathbb{E}_x \prod_{j=1}^q g(x + y_j).$$

Conditional on the proposed positive-product estimates, this suggested cancellation for grids with one long axis. Rare structured holes nevertheless defeated stronger moment bounds: taking a large root could erase the benefit of their small probability. Weighted Cauchy–Schwarz preserved useful support restrictions only until the weights were replaced by approximations; completion weights could vanish exactly where μ was supported. The relative Szemerédi framework of Conlon, Fox, and Zhao [7] suggested a transfer route, but the assistant found that the needed lower estimates for linear forms reintroduced the counting problem it was trying to solve.

The exploration then turned to an ideal jointly high-rank quadratic factor. Its four-term relation produced Fourier terms $|\hat{f}(\xi)|^2 |\hat{f}(3\xi)|^2$ for a common real-valued factor function f , suggesting positivity:

VERBATIM EXCERPT

| Good! Degree2 polynomial structure *doesn't obstruct same-set 4AP lower* when high-rank jointly.
| Obstruction roots in low-rank/lower-degree factors.

The assistant next seeks quantitative estimates for that ideal model. A subsequent spectral-growth argument encountered a product-set counterexample, prompting an additional stability requirement under conditioning. Entropy-based extraction still lost too much density. The assistant was still seeking a lower counting theorem and a transfer from the finite-field model to integers.

5 From rare holes to a conditional convexity gain

The assistant's sparse progression-counting arguments repeatedly encountered one obstruction: upper bounds on concentration do not stop a sparse set from occupying rare holes in a completion function. Proposed entropy projections sought to remove a small structured subspace; padding sought a non-negative majorant with two-sided pseudorandomness. The padding proposal invoked Conlon–Gowers transference [8]; the relative Szemerédi argument of Conlon, Fox, and Zhao [7] sharpened the concern that an upper linear-forms bound alone would not fill these holes. The assistant did not complete either construction. The assistant also considered Rankin's progression-free constructions as possible counterexamples [47], whose known density scale left a large gap to the desired logarithmic bound. A decision-tree argument looked promising for complements of disjoint affine subspaces, but overlapping weighted constraints remained unresolved. Yannakakis's clique-versus-stable-set communication argument [52] suggested a related way to split disjoint rectangles; extending it to higher-order cylinder intersections remained uncertain.

The spectral approach similarly recovered only a cubic singular-value bound. Gaussianizing correlated directions consumed additional entropy along with the apparent gain. Hypergraph reformulations exposed dependent links, and controlling positive moments of a completion function still failed to control its zeros where the target weight concentrated:

VERBATIM EXCERPT

| Need lower tail at μ_H . Upper moments no.

Redundant arithmetic witnesses and Croot–Sisask almost-periodicity [9] were then considered as ways to fill those holes. Revisiting the graph-counting mechanism of Filmus, Hatami, Hosseini, and Kelman [12] highlighted the difficulty of replacing graph common neighborhoods by cyclic hypergraph links. A proposed sumset of missing endpoints required Cartesian families of partial progressions too large to expect even in random sparse sets.

The assistant next used independent shifts of a normalized density μ , with $g = \mu - 1$. Expanding $\prod_j \mu(x + u_j)$ into products of shifted copies of g , the assistant proposed applying uniform convexity in Fourier ℓ^3 . The Kelley–Meka method and Bloom–Sisask exposition [33, 4] motivated the positive-product bound needed here. Conditional on that bound, exchangeability would supply an inverse binomial factor and hence exponential decay for centered products. The assistant reacts:

VERBATIM EXCERPT

| This solves $k=4$ counting lemma via Holder! Also arbitrary matrices S_3 via cotype martingale? Need positive upper binomial-type $\ast\text{norm}\ast$ for patterns.

It next examines the positive-product bound needed for this gain. Extending the argument to longer progressions generated shift grids whose cycles destroyed independence. Inside a quadratic atom B , centering also had to use $1_B / \text{dens}(B)$, introducing costly density factors. Local Fourier norms and a relative Bessel inequality suggested repairs, but the resulting ambient transfer could be signed. The assistant also revisited the polynomial analytic-rank-to-partition-rank bounds of Janzer and Milićević [29, 44] as a possible route from bias to lower-degree structure. The bounds of Leng, Sah, and Sawhney [35] remained a quantitative comparison; their inverse theorem and Manners’s cyclic-group inverse theorem [36, 41] were considered as inputs, with the required finite-field analogue still uncertain. The assistant was left seeking positivity, spread, controlled iteration, and a passage to the integers.

6 From random restrictions to the cancellation barrier

For a nonnegative function of mean one bounded by e^p , the target was a positive progression count with losses at most exponential in a polynomial in p . The working assumption excluded density increments on factors generated by relatively few polynomial tests. The assistant kept Green–Tao’s four-term bound and the Leng–Sah–Sawhney bounds and inverse theorem as quantitative benchmarks, while questioning the finite-field inverse input needed by its proposed transfer [17, 35, 36].

Building on its own preceding proposals, the assistant first sampled affine subspaces contained in a polynomial atom. Restricting to these spaces preserved progression avoidance, but transferring local structure back to the ambient space threatened excessive complexity. The assistant then tries Hahn–Banach separation: measure the residual by its average restricted phase correlation, and require boundedness only of the separating functional. It observes:

VERBATIM EXCERPT

| no need residual cap except in dual where bounded. Excellent!

The proposed decomposition still lacked an affordable iteration. Counting also produced repeated-point contributions that could overwhelm the progression signal.

The convolution estimates continued to draw on Kelley–Meka sifting, Bloom–Sisask’s exposition, and Croot–Sisask almost-periodicity [33, 4, 9]. The assistant also noted that the linear-forms hypotheses in Conlon–Fox–Zhao’s relative theorem supplied more than its upper-only spread assumption [7].

Attention shifted to higher-order norms. Fixed cube norms appeared to undercount many independent polynomial modes, while repeated Hölder inequalities amplified rare structured holes. The next proposal replaced a potentially exponential spectral-energy budget by an entropy budget of order p . Independent structured projections could share that smaller budget; dependencies among them prevented a completed argument.

The final stage tried polynomial spectral models and penalized least-squares approximations. Matching moments did not preserve positivity. Approximating progression dual functions directly in L^2 , through bias and the partition-rank bounds of Janzer and Milićević [29, 44], led it to a further difficulty: grouping nearby phases could destroy cancellations. Small partition-boundary probabilities became inadequate after multiplication by large coefficient sums:

VERBATIM EXCERPT

| Need a norm-based operator bound for Schur multiplier error rather than entry absolute.

The remaining target was sharper control of four-factor Fourier sums through energies on low-dimensional affine subspaces. The assistant had not resolved the cancellation problem: its proposed route still required an affordable decomposition, control of repeated-point contributions, and a norm-based estimate that survived grouping phases. It therefore left the finite-field counting target and the subsequent transfer to integers as further steps.

It also reconsidered the binary-system counting theorem of Filmus, Hatami, Hosseini and Kelman and the corners bounds of Jaber, Liu, Lovett, Ostuni and Sawhney as possible reductions [12, 28]. Its proposed lifts still failed to force a longer collinear progression: a corner supplied a parallelogram, and the simple-graph representation encountered a rank obstruction.

7 From adaptive holes to compressed polynomial constraints

The assistant sought lower bounds for longer progression counts from upper bounds on conditional densities. For a set of density $\alpha = e^{-p}$, its normalized indicator $\mu = 1_A/\alpha$ has mean one but can reach e^p . This large cap made rare exceptional holes decisive: estimates adequate for ordinary averages could fail when tested against the sparse set itself.

Kelley–Meka moment estimates remained a model for controlling convolution and Fourier energy [33]. The assistant also considered Sanders’s small-doubling structure [48], but did not obtain the desired small-codimension control from it. Adaptive Fourier projections briefly offered a way to avoid enumerating all structured factors. An empirical Fourier multiplier controlled projected energy in expectation, but the required exponentially small tails remained unavailable. Hypergraph containers needed too many moments; the Conlon–Fox–Zhao densification framework [7] risked assuming the lower count it was supposed to prove. Croot–Sisask almost-periodicity [9] suggested sampling shifted completions, but preserving a deficit on a sparse test set required more accuracy than the assistant could supply. Weighted Cauchy–Schwarz produced facet-weighted cubes, yet their needed high-moment estimates recreated the original difficulty. Spectral entropy and martingale proposals likewise failed to prevent complexity or entropy from concentrating in unfavorable places. The assistant explored the global hypercontractivity of Keevash, Lifshitz, Long, and Minzer [30] as a source of Fourier level bounds; concentrated dependencies and high levels still obstructed the proposed counting argument. It compared the goal with Green–Tao’s polylogarithmic four-term bound [17] and Leng–Sah–Sawhney’s longer-progression bounds [35], without finding an amplification that closed the gap.

The assistant next asks how to iterate a finite-field increment, assuming it can obtain the increment lemma. In an atom defined by polynomial equations, aligning local phases introduced many lower-degree corrections. The proposed repair retained the original equations as expressions in auxiliary variables, then averaged substitutions for discarded directions. Processing variables by degree might preserve both polynomial degree and the number of constraints.

Nested affine flats were then proposed to separate rank requirements by degree. The assistant tried to use Lampert–Ziegler relative rank and regularization [34] to make each layer’s rank requirement depend on that layer and the higher ones. It invoked the polynomial analytic-rank/partition-rank comparisons of Janzer and Milićević [29, 44] when turning biased polynomial combinations into lower-degree pieces. The quantitative inverse input remained uncertain: it recalled the Gowers–Milićević finite-field U^4 theorem [14], contrasted finite-field needs with the Leng–Sah–Sawhney integer inverse theorem [36], and still needed the appropriate hypotheses and bounds for the proposed iteration.

Finally, summing lifted field coordinates suggested that integer carries might cancel with polynomial rather than exponential probability loss. Its image was too small, and enlarging it left collapsed progressions and carry constraints unresolved. The assistant ended with the increment theorem still assumed and with unresolved steps in both its iteration and the transfer to integers.

8 From slab collapse to entropy counting

The assistant moved from finite-field embeddings toward an entropy-based counting principle for integer arithmetic progressions, motivated by its earlier candidate four-term argument.

The embedding proposal restricted sums of canonical coordinate representatives to intervals of width less than half the field size. This forced their second differences to vanish, so vector progressions mapped to integer progressions. But nonzero vector directions could lie in the kernel of every block sum, producing constant images. Density increments on simple residue classes reinforced this collapse:

VERBATIM EXCERPT

Finite-field linear tests fully expose residue at small cost ($\log q$ bits), so reduction fails structurally, not just counts.

The assistant revisited the Leng–Sah–Sawhney density-increment and inverse-theorem inputs [35, 36], while continuing to question the quantitative finite-field inverse statement its proposed transfer would require. It also retained the limitation of the binary-system counting theorem: the simple-graph encoding did not directly handle longer progressions [12]. Its polynomial-factor calculations used the polynomial analytic-rank/partition-rank bounds of Janzer and Milićević [29, 44]; the relative-rank approach of Lampert and Ziegler remained an input whose precise applicability it wanted to check [34].

Attention returned to normalized sparse densities bounded by $M = e^p$. Upper spread constrained their averages on structured sets, but could not directly exclude a completion function vanishing on the entire target set. The exploration reused Kelley–Meka convolution and sifting ideas, including the almost-periodicity formulation in Bloom–Sisask’s exposition [33, 4], and considered Croot–Sisask smoothing of partial-completion functions [9]. Positive covering certificates, weighted localization, and phase-based norms were explored. Repeated Hölder amplification created correlated grids, whose aligned holes defeated the hoped-for independent cancellation; subgroup-block examples also defeated a proposed spectral comparison.

The assistant contrasted its upper-only hypotheses with the linear-forms conditions used by Conlon, Fox, and Zhao’s relative Szemerédi theorem [7]; the missing lower estimates blocked a direct densification argument. An abstract replacement asked whether normalized simplex-facet kernels, with uniform proper marginals and upper control on large cylinder intersections, must have positive joint count. The

assistant did not find a proof of this lower bound or a derivation of its hypothesis from polynomial spread.

A coordinate-filtration approach then exploited pairwise independence: surviving progression corrections involved at least three innovations. Absolute innovation estimates nevertheless counted irrelevant late spikes:

VERBATIM EXCERPT

Thus simple martingale fails. Maybe choose **adaptive basis via polynomial inverse** such that relevant frequencies revealed early before spikes materialize.

A proposed comparison retaining two arbitrary weighted marginals might charge each correction to $e_j^{3/2}$, with $\sum_j e_j \lesssim p$. Small steps $e_j \leq t$ would then cost $O(p\sqrt{t})$. Large steps remained dependent on enormous previously revealed prefixes, and transferring them recursively escalated complexity. It turns to compressing the previously revealed prefixes while preserving the proposed comparison.

9 From entropy jumps to positive grids and integer obstacles

The assistant sought progression-count estimates for a normalized density $\mu \leq M = e^p$. Entropy increments along finite-field coordinates summed to at most p , suggesting that sufficiently small increments might have a controllable total effect. The assistant considered transference through the linear-forms conditions of Conlon, Fox and Zhao [7], but large revelations resisted this argument: repeated transference magnified errors, and backward coordinate deletion failed because parity-type dependencies could force every coordinate to remain.

Random subspaces and graph-label formulations concentrated the difficulty but introduced too many Fourier modes or left a new counting problem. It also considered the same-set hitting framework of Hązła, Holenstein and Mossel [26], observing that exact progression relations fell outside the straightforward product-space route. Continuous revelation did not remove the obstruction: pairwise-independent endpoints can have zero joint product, whereas bounded continuous orthogonal martingales would preserve its expectation.

The direction changed when correlated affine functions produced individually nonnegative covariance kernels,

$$\mathbb{E}_\sigma v_i(u)v_i(v) = 1 - t + tuv, \quad 0 < t < \frac{1}{2},$$

while their cross kernel had a negative uv -coefficient. A deficit in the full progression count could consequently become an excess in a mixed average.

The assistant then identified what this proposal still required. It proposed Hölder and Cauchy-Schwarz bounds for the excess through positive additive-grid averages, but still needed the grid bounds or a structural consequence of their failure.

The next stage made that dependency explicit:

VERBATIM EXCERPT

This is big. Need convert single-grid excess to density increment over integers with manageable complexity, avoid previous integer relative transfer burden.

Isolating a grid cell suggested a positive correlation against functions of proper subsums. Proposed integer conversions used nilsequences, multidimensional boxes, or direct comparison with a structured baseline. Repeated interval restrictions accumulated excessive losses. Large boxes might reduce degeneracy, but conditioning and lifting remained unresolved; Bohr neighborhoods introduced exponential

geometric normalization losses, while coefficient heights could grow despite dimension control. The assistant continues to seek an integer density increment from the kernel reduction.

In pursuing the integer conversion, the assistant returned to Kelley–Meka sifting, Bloom–Sisask’s exposition, and Croot–Sisask almost-periodicity as models for controlling sparse convolutions [33, 4, 9]. It recalled the three-term reciprocal-sum consequence of Bloom–Sisask and the polylogarithmic four-term bound of Green–Tao [3, 17], while keeping the graph-specific scope of the binary-systems result of Filmus, Hatami, Hosseini and Kelman in view [12]. For longer progressions it examined the nilsequence inverse theorem and density-increment argument of Leng, Sah and Sawhney [36, 35]: repeatedly localizing to intervals still accumulated excessive losses.

For a possible finite-field transfer, it proposed using the polynomial analytic-rank-to-partition-rank bounds of Janzer and Milićević [29, 44] to reduce biased polynomial combinations to lower degrees, and Lampert–Ziegler relative rank and regularization [34] to control successive constraint layers. It treated the required sampling and complexity estimates as dependencies still to check. For the integer analogue it identified Leng’s efficient equidistribution of nilsequences [37] as a possible source of a degree-reduction step, but did not establish that the needed form followed.

10 Degree reduction and alternating local decompositions

If $r_k(N)$ is the largest size of a subset of $[N]$ without a nonconstant k -term arithmetic progression, the target was $r_k(N) \ll N/(\log N)^2$. The assistant combined the Leng–Sah–Sawhney inverse theorem with their progression-decomposition strategy [36, 35]. With $p = \log(1/\alpha)$ for density α , repeated restrictions had to consume only polynomial complexity in p .

The assistant examined Leng’s quantitative nilsequence equidistribution results [37], but found that their horizontal-character obstructions did not supply the required degree reduction: lowering nilpotent step can leave polynomial degree unchanged. It also reconsidered the step-lowering mechanism for periodic nilsequences [38]. The assistant proposed a suspension:

VERBATIM EXCERPT

Key idea! Linearize polynomial nilsequence via suspension to linear sequence in s -step nilmanifold with standard lower central series, preserving dimension and complexity.

The assistant then notices that an integral drift can disappear modulo the lattice, defeating its proposed subgroup restriction. Derivative representations offered another route, yet dividing by a small phase mean incurred prohibitive costs.

Green–Tao local quadratic inverse methods and their polylogarithmic four-term progression argument suggested working in local charts [18, 17]. Green–Tao–Ziegler nilcharacter calculus then informed the proposed derivative representations [22].

Attention shifted to domination against positive lower-degree tests and then to a convolution comparison with an arbitrary nonnegative baseline. Simultaneous derivative bias and anchoring suggested alignment mechanisms, but closure under translated weights remained unproved. Variance also failed to guarantee useful progress when concentrated on rare spikes. Attempts to construct convolution predictors used Croot–Sisask sampling and Sanders’s Bogolyubov–Ruzsa mechanism, but encountered label-entropy and difference-set obstacles [9, 48].

The proposed repair alternated independent Bohr-set pieces, used to remove spikes, with matched pieces, used to detect convolution nonuniformity. A concave potential $(hw)^\theta$, where h, w are local means and $0 < \theta < 1/2$, was intended to control dependent refinements and unfinished branches. Kelley–Meka autocorrelation estimates and dependent random choice supplied the model for variance incre-

ments [33]. The assistant examined the finite-field and local Bohr almost-periodicity statements in Bloom–Sisask, including their Schoen–Sisask input, as the analytic bridge [4, 49]. The assistant accepted the local comparison at this stage, but repeated conditioning on structured atoms could still multiply complexity. Flattening existing phases incurred excessive losses; a hierarchy of local coordinates became the next proposal. The assistant left the comparison proof and the higher-degree iteration unfinished.

Reconsidering the finite-field comparison also brought back the Janzer and Milićević polynomial rank bounds and Lampert–Ziegler relative regularization as possible controls on nested sampling [29, 44, 34]. The assistant still needed to check their quantitative dependencies. It also retained the earlier obstacle to applying Conlon–Fox–Zhao relative counting: the required control of linear forms in the envelopes was missing [7].

11 Keeping relative density increments affordable

The assistant next considers the cost of repeating density increments. If $r_k(N)$ denotes the largest size of a subset of $[N]$ without a nonconstant k -term arithmetic progression, the desired bound was $r_k(N) \ll N/(\log N)^2$. With relative density $\alpha = e^{-p}$, repeatedly replacing structural dimension D by D^C would consume the available scale. Building on its own earlier proposals, the assistant sought new complexity controlled by relative density rather than the rarity of an existing structured region B :

VERBATIM EXCERPT

Need relative inverse theorem *increment count* depending only relative density*, rather than black-box HB paying b .

Here HB denotes Hahn–Banach decomposition. The assistant used the inverse-theorem framework of Leng, Sah, and Sawhney [36] as background and sought a relative extension.

The principal construction sampled affine boxes $\psi(t) = x + v \cdot t$ on which B was nearly constant. A local test into a small nilmanifold would then have to become a global test into the same target. Interpolation alone could not preserve correlation: a character nearly constant on a small box might average to zero globally. Conditioning boxes to stay inside B could also make the sampling event too rare. Encoding finite-field progressions in integers brought separate carry and collapsed-direction obstructions.

The finite-field comparison invoked Lampert–Ziegler relative rank and regularization [34], while retaining uncertainty about the precise regular-tower hypotheses, and Janzer’s polynomial comparison of partition and analytic rank [29]. For nilsequences, the assistant examined Leng’s periodic and multiparameter equidistribution results [38, 37], hoping to lower degree without repeated dimension-dependent factorization. It contrasted this with the cost of quantitative Leibman factorization [23] and recalled the nilcharacter-symbol framework of Green, Tao, and Ziegler [22]; the required compatibility of the lifted coefficients remained a proposed step.

The assistant proposes representing B as a product of smooth tests. If every added test retained sufficient relative mass and stable output size, logarithmic mass and complexity costs could accumulate additively. It therefore asks for a transfer theorem supplying those hypotheses. Ordinary polynomial phases offered a simpler setting for coefficient matching before confronting nilpotent structure.

The assistant used the Schmidt-type recurrence and progression decomposition of Leng, Sah, and Sawhney [35] to estimate neighborhood mass and the losses from repeated localization. Green–Tao’s local U^3 inverse theorem [18] suggested a model for adding structure on Bohr sets, and Manners’s periodicization construction [42] was considered for cyclic versions. Recalled comparison arguments combined Kelley–Meka sifting [33], Croot–Sisask convolution smoothing [9], and the Bohr-set almost-periodicity theorem in Bloom–Sisask’s exposition [4]. The assistant also retained its objection to the

Conlon–Fox–Zhao transference route [7]: the required linear-forms control was precisely what its sparse argument had not supplied.

The assistant then asks to preserve only correlations needed for one density increment. A variance estimate of order L^{-m} still could not control a candidate family of size $L^{O(dm^s)}$ for degree- s maps:

VERBATIM EXCERPT

However to correlate with $f \circ \psi$, h must align with f model (low entropy) in top tensor space, constraints leave free components not needed (prune).

This suggested discarding irrelevant coefficients and lifting the remaining coordinates layer by layer while respecting commutators. The assistant left the quantitative lifting theorem and the full iteration as tasks still to be completed.

12 Why controlling dimension does not control width

The assistant sought a density increment inside a structured region B . With density $\alpha = e^{-p}$, the goal was to add only $\text{poly}(p)$ coordinates per increment while preventing their required precision from overwhelming the iteration.

Local nilsequence tests were first to be lifted from affine boxes to the integers. The assistant considered Leng’s efficient equidistribution and periodic step-reduction results, while questioning whether their hypotheses covered the required multiparameter boxes [37, 38]. It also tested a recollection of high-characteristic nilspace splitting against explicit periodic Heisenberg examples [6]. A proposed simplification used global coefficient lifts and one random constant in a projection kernel to reproduce a joint distribution. Incompatible lattice gauges still obstructed simultaneous lifting. Using Leng–Sah–Sawhney inverse estimates and progression localization as quantitative inputs [36, 35], the assistant obtained a conditional budget calculation by grouping increments before resetting the structured region, but nonlinear dependence on previous precision could defeat it.

The assistant revisited Kelley–Meka sifting and Bloom–Sisask almost-periodicity in its proposed positive convolution comparison [33, 4], and recalled Green–Tao’s locally quadratic approach to four-term bounds as another model [17]. The next route conditioned progression counts on B , generating additive grids through repeated Hölder inequalities. Their final two positions remained coupled. A bounded coupling density could favor positive inner products even when the separate Gram kernels were non-negative, invalidating the hoped-for independence substitute. Taking absolute values erased the useful negative contribution.

Ordinary polynomial models suggested absorbing some lifted frequencies into existing coordinates. Later, selecting independent frequency equations gave a proposed pointwise alignment between a local test, a global polynomial, and a slow drift. This simplified algebraic compatibility without controlling the drift’s oscillation. Fixed weighted degree likewise suggested polynomial dimension growth, yet hidden floor variables and localization costs remained unresolved.

A proposed relative inverse principle also raised the question whether Conlon–Fox–Zhao transference and its linear-forms hypotheses could remove losses depending on the width [7]. Sampling whole atoms, traversing short discrete labels with long kernel directions, and using high-dimensional boxes each encountered further obstacles. A positive quadratic phase, modeled by $\|u\|^2/L$, forced substantial shortening:

VERBATIM EXCERPT

Positivity smooth irrational low-rank counterexample forces log cost factor 2 independent d !

The assistant then pulled the integer set back from a finite-field affine space, where kernel directions create degenerate progressions. Their proportion alone did not settle the necessary comparison with progression bounds. The assistant was still seeking a width-stable increment theorem, with the finite-field comparison unresolved.

13 Amplifying density gains without multiplying dimension

With density α and $p = \log(1/\alpha)$, the obstacle was the expense of conditioning on an increasingly narrow structured region: inverse-theorem accuracy and subsequent localization depended too strongly on that region's tiny mass.

The quantitative inverse and progression-decomposition estimates of Leng–Sah–Sawhney framed this cost problem [36, 35]. Seeking cheaper inverse estimates, the assistant recalled the polynomial Freiman–Ruzsa results of Gowers, Green, Manners and Tao in characteristic two and bounded torsion, but did not obtain the needed integer-group analogue [15, 16].

The proposed escape was to increase the gain per iteration. Several small increments might form a packet giving a large fixed factor K .

VERBATIM EXCERPT

Then multiplicative increment factor K per iteration, choose K huge enough to absorb block constants! This is extremely important.

The assistant distinguishes coordinate count from precision. Widths might deteriorate rapidly, provided the exponent controlling their recurrence remained independent of K , and the number of coordinates stayed polynomial with a similarly uniform exponent. The assistant needed a packet lemma with these uniform exponents for the parameter calculation to apply.

One route toward a larger gain sought a bounded model: a suitable positive majorant would permit Conlon–Fox–Zhao relative Szemerédi transference [7], but the proposed domination argument encountered rare overlapping tests. A second route considered the Lovász–Szegedy graph-limit characterization through homomorphism densities and reflection positivity [40]; the assistant could not obtain the mixed-sign positivity or counting transfer needed for its sparse weights.

Attempts to lift local tests globally exposed scale feedback. Short coordinates became frozen labels; accounting for them demanded greater approximation accuracy and hence larger sampling boxes, which encountered further scales. Enlarging an ambient cyclic group created many apparent directions but only a short effective progression in the original integers. Finite-field and product models likewise left unresolved whether a detected progression would collapse to one point or incur excessive losses when controlling carries. A direct, density-dependent large increment also failed its cost comparison: the next test's complexity outgrew the gain.

For finite-field regularization, the assistant revisited a construction conditional on Lampert–Ziegler relative rank estimates, using Janzer's analytic-rank to partition-rank bounds in its proposed rank checks [34, 29]. On the integer side, it considered Green–Tao's local quadratic inverse theorem and their polylogarithmic four-term progression bound, while questioning whether the local inverse costs could fit its logarithmic-density budget [18, 17]. Leng's periodic and multiparameter nilsequence equidistribution results remained candidates for aligning local tests, with their precise lifting application still unresolved [38, 37].

The exploration returned to conditional counting, using Gaussian weights or polynomial coefficient data to seek replication costs depending on p , rather than the conditioning mass. The underlying comparison proposal combined Kelley–Meka variance and sifting ideas with Bloom–Sisask Bohr almost-periodicity [33, 4]. The remaining task was to control the number of new structural coordinates in the comparison and inverse steps.

The assistant thus focused the relative lifting problem on uniform coordinate bounds, while retaining its doubts about the absolute comparison argument on which the proposed density iteration depended.

14 Controlling dimension through repeated density increments

The assistant sought a density-increment proof of Erdős’s reciprocal-sum conjecture. Its central difficulty was quantitative: structure detected inside an already constrained region had to be transferred to the ambient integers without making the number of coordinates grow too quickly. Writing $p = \log(1/\alpha)$ for density α , repeated updates $d \mapsto d^C$ could exhaust the dimension budget long before enough fixed-factor density gains accumulated.

The assistant tried conditioning polynomial Taylor coefficients to construct a sampling law, then identified uncontrolled sparse marginals and exceptional fibers. Linear Bohr regions admitted progression charts; higher-degree regions did not have comparable affine charts with the required cube distributions. Finite-field encodings introduced wraparound and carry losses, while grouping increments into packets still retained excessive dimension growth.

The dimension calculations revisited the inverse-theorem and progression-decomposition bounds of Leng–Sah–Sawhney [36, 35]. For the convolution stage, the assistant considered Kelley–Meka concentration arguments [33], Bloom–Sisask almost-periodicity [4], Croot–Sisask smoothing [9], and Sanders’s Bogolyubov–Ruzsa lemma [48]; the difficulty was keeping their costs under control inside the old constraints. It also reconsidered a bounded-majorant route to the Conlon–Fox–Zhao relative Szemerédi theorem [7], but the needed two-sided control was missing.

Green–Tao’s polylogarithmic four-term bound suggested examining local quadratic methods [17]. The assistant considered whether the binary-system counting method of Filmus, Hatami, Hosseini, and Kelman could bypass higher-order structure, but could not encode the required longer progression that way [12]. It also recalled possible distinct-degree polynomial methods without settling their bounds or a reduction to ordinary progressions. In the finite-field model, it revisited relative-rank regularization in the style of Lampert–Ziegler, leaving the needed sampling and threshold dependencies open [34].

Polynomial charts led to a more concrete idea for short coordinate directions. For an ordinary degree- r polynomial, comparison with a fixed short-coordinate slice gives

$$P(t) - P(t_{\text{long}}, a) = \sum_j (t_j - a_j) D_j(t), \quad \deg D_j \leq r - 1.$$

The assistant proposed descending through these lower-degree terms:

VERBATIM EXCERPT

For degree $r \geq 2$, if difference has fast variations depending short axes, can express via lower-degree polynomials t_j and mixed terms without Weyl!

The assistant then encountered further obstacles in extending this ordinary-polynomial proposal to nilsequences. Rational and smooth corrections introduced floor coordinates whose number was still uncontrolled. The assistant considered Leng’s periodic nilsequence step reduction [38] and reread the multiparameter equidistribution estimate [37], observing that its constants depended on the number of parameters. Separating quadratic coefficients into oscillatory and structured components led to circular precision requirements.

The absolute comparison remained unverified.

A subsequent proposal maintained a high-dimensional unconstrained box, spending dimension to preserve side length. The assistant found that its initial encoding cost exceeded the budget. It then tried

noninjective affine maps, but observed that localized progression directions could all lie in the map's kernel. Ensuring nonzero image directions reintroduced the missing relative sampler. The assistant next considered prime-power digits and entropy, while still seeking the relative sampler needed to continue the density-increment argument.

15 From relative smoothing to positive sampling laws

The assistant shifted from relative counting to transferring local gains, seeking density increments strong enough for Erdős's reciprocal-sum conjecture. Hilbert cubes did not force arithmetic progressions, and a replicated-grid calculation left a small but consequential excess exponent. The assistant tried asymmetric Gaussian profiles and retained support indicators, then sought comparison estimates for them.

Before this smoothing proposal, the assistant considered Green–Tao transference through a pseudorandom majorant, but did not find a suitable majorant for an arbitrary progression-free set [20]. It also considered Conlon–Fox–Zhao densification [7] and a dense-model argument; the required accuracy appeared to demand excessive pseudorandomness. Kelley–Meka sifting and the Bloom–Sisask almost-periodicity theorem supplied the comparison framework [33, 4].

A proposed relative adaptation of Croot–Sisask sampling [9] then used reference-measure autocorrelations and Fourier uniformity to seek smoothing costs independent of sparse support density. The assistant proposed the following local analogue:

VERBATIM EXCERPT

For local Bohr, similar using relative convolution and locally embedded group with support. This yields relative Croot-Sisask almost-periodicity under mere β autocorr and β' Fourier. Major progress.

For finite-field sampling, the assistant revisited Lampert–Ziegler relative-rank regularization and Janzer's rank bounds, while explicitly questioning the required uniformity in the earlier layers [34, 29]. It considered local quadratic inverse arguments and Sanders's local Bogolyubov–Ruzsa approach to controlling new directions. Green–Tao's polylogarithmic four-term bound suggested a separate amplification route, but tensor products and fiber bounds did not supply the desired improvement [17]. The weaker bound for longer progressions remained a benchmark [35].

The assistant still needed a local argument and turned to retaining fixed widths for quadratic constraints while shrinking their effective rational subspace. This was intended to prevent repeated coordinate elimination from multiplying precision costs. For structured approximation it continued to use the Leng–Sah–Sawhney inverse theorem [36]. It examined Leng's multiparameter nilsequence estimates, but did not assume that they supplied the stronger degree-reduction statement or the dependence on the number of parameters that it needed [37]. A lattice-based alignment proposal sought a decomposition of horizontal coordinates into isotropic, rational, and slowly varying components. The assistant questioned how to extend that quadratic mechanism to higher degrees.

The next obstacle was sampling: an affine chart could overweight ambient points, invalidating separation estimates taken with uniform ambient measure. Truncating exceptional sampling weights required bounds independent of chart length. Pair concentration also encountered congruence correlations and circular choices of sampling dimension. Studying rational characters annihilating the chart's slope matrix suggested a common denominator in place of many unrelated moduli:

VERBATIM EXCERPT

This implies finite group generated by needed annihilators has exponent $< D^0$ (or $|\text{group}| \leq D^{0^n}$), eliminating lcm! This is key.

That proposed typicality argument motivated a positive ambient weighting reproducing a chart's position, residue class, and polynomial labels. Including such weightings in the approximation norm might transfer a local gain under the assumption that no ambient density increment exists. The assistant left uniform Fourier bounds, independence from polynomial constraints, smoothing of lattice labels, and consistent parameter ordering as unresolved requirements for this transfer proposal.

16 Controlling dimension through graded lifting

The assistant addressed the dimension growth obstructing a density-increment approach to Erdős's reciprocal-sum conjecture. The desired conclusion was progressions of every finite length. Motivated by Leng's periodic and multiparameter nilsequence estimates [38, 37], it sought a relative lifting theorem.

For density α , put $p = \log(1/\alpha)$. Grouping increments into a fixed multiplicative gain could tolerate narrower sampling regions, but required structural dimension polynomial in p . A recurrence $d_{i+1} \leq d_i^C p^C$ defeated this strategy. Moreover, replacing a small local nilsequence by a large ambient comparison sequence imported hidden integer-part coordinates. The construction still had to cover every fixed progression length, rather than only four-term progressions.

The assistant revisited several ingredients behind the proposed iteration. Kelley–Meka concentration methods and Bloom–Sisask almost-periodicity supported its degree-one comparison strategy [33, 4]; it also explored a weighted adaptation of Croot–Sisask sampling [9]. It considered applying the relative Szemerédi theorem of Conlon, Fox, and Zhao after constructing a suitable majorant, but still lacked the required two-sided control [7]. For a finite-field alternative, it relied conditionally on Lampert–Ziegler relative-rank regularization and Janzer's polynomial relation between partition and analytic rank [34, 29]. Green–Tao's local inverse approach to four-term progressions suggested another route, whose quantitative dependence remained an obstacle [17]. The Leng–Sah–Sawhney inverse theorem supplied candidate structured models, while their progression decomposition supplied the proposed reset to ordinary progressions [36, 35].

Alternative routes through overlapping shorter progressions, Chinese-remainder independence, and coordinate fibers produced no replacement argument. Bracket phases retained troublesome floor interactions, and fiberwise bounds did not automatically multiply. Attention returned to a graded factorization that would iterate over degree rather than ambient dimension. The assistant examined the dimension losses in Green–Tao factorization [23] and Leng's comparison with the sunflower and linearization steps in Green–Tao–Ziegler's U^4 inverse argument [37, 24]. It noted that Leng's multiparameter theorem allowed its exponent to depend on the number of parameters, leaving a further uniformity question. Smooth and rational factors were to leave a subgroup annihilated in its highest layer by a vertical character. The assistant tried coefficientwise adjustments to improve compatibility, then returned to the derivative character and its quantitative bounds.

Conditioning on existing constraints introduced weighted floor variables. It briefly considered nilprogression-based dimension control for subgroup charts. A semidirect polynomial model suggested rational potentials for these corrections; pruning Fourier modes before freezing short sampling axes was intended to preserve useful correlation. To prevent recursive corrections from accumulating coordinates, the assistant proposed:

VERBATIM EXCERPT

Better: align entire map g_P (auxiliary okay large) into SAME via iterative Fourier pruning and simultaneous splittings, preserving local tests as modifications of one map into original target with Y fixed.

Here Y denotes fixed ambient data. Quotienting a compatible central subgroup and reconstructing the original target offered a more specific lifting scheme. The assistant still needed to resolve lattice compatibility, well-defined actions, global bias, and efficient factorization to complete this lifting scheme.

17 From graded regularization to a structured comparison problem

The density-increment strategy for Erdős's reciprocal-sum conjecture shifted from controlling polynomial complexity to confronting arithmetic sampling obstructions. The assistant had proposed lifting local tests while preserving marked projections. A graded revision would transfer a deficient top-degree component to lower degree, bounding the total number of reductions. Integer-coordinate compression was intended to limit the number of floor variables, but the assistant had not resolved its interaction with boundaries and density.

The assistant revisited the finite-field regularization route using Lampert–Ziegler's relative rank and Janzer's analytic-to-partition-rank bounds [34, 29]. For the integer problem, it contrasted Green–Tao's local inverse argument [17] with the quasipolynomial inverse theorem of Leng–Sah–Sawhney [36], while retaining the latter authors' effective recurrence/localization input for a later return to a one-dimensional progression [35]. Leng's periodic nilsequence estimates remained a proposed source of subgroup identities [38]; the assistant also recalled that the multiparameter theorem's exponent depended on the number of parameters [37].

The sampler had to accommodate both short and long lattice coordinates. Choosing its length only after regularization appeared to replace an expensive power separation by a multiplicative gap:

VERBATIM EXCERPT

This previous failure was *power gap* $L^{\{\text{poly}\}}$ requiring $\log L$ multiplying D times $\rightarrow$ tower. Here gap only factor A independent L !

The assistant noticed that narrow coefficient jitter could leave discrete labels sparse and that real smoothing lost rational oscillations. The proposed comparison weight therefore combined continuous information with compatible congruence residues, seeking relative rather than absolute approximation for tests of very small mass. A further simplification allowed the final density-increment region to replace the old region, potentially reducing inherited complexity.

The assistant then encountered a dimensional difficulty: an affine sampler with $\ell < n$ directions has a singular image modulo every prime in an n -dimensional ambient box. In an ideal Chinese remainder model, orthogonal decompositions offered useful contraction estimates, but too many singleton-prime contributions survived. The assistant proposed using global hypercontractivity [30], later identifying the sharp inequalities of Keller–Lifshitz–Marcus [32] as a possible ingredient, without yet deriving the needed comparison estimate. It also reconsidered Kelley–Meka's weighted energy and sifting arguments [33], together with the finite-field and Bohr-set almost-periodicity statements in Bloom–Sisask [4], as models for adaptive comparison.

A counterexample then concentrated an arbitrary baseline on a union of rare congruence events. Small restrictions left many events unseen, while a typical low-dimensional sampler missed the union.

The proposed repair used the actual baseline's bounded-complexity nilsequence structure. Repeated congruence sensitivity might force rational constraints, eventually producing stable averages. The assis-

tant left three steps to establish: this regularization, the hypercontractive comparison, and transfer from ideal prime independence to finite boxes.

18 From physical residue mixing to symbol regularization

Continuing the attempt at Erdős's reciprocal-sum conjecture, the assistant sought quantitative bounds on progression-free sets strong enough to force arithmetic progressions of every length. Several lemmas in the proposed density-increment plan remained unverified.

The assistant recalled the contrast between Green and Tao's polylogarithmic four-term bound [17] and the longer-progression estimates, and the restriction that had prevented it from directly applying the sparse graph-counting result of Filmus, Hatami, Hosseini, and Kelman [12] to longer progressions.

The immediate challenge was transferring averages from a low-dimensional affine sample $\psi(t) = x + Vt$ to a larger integer box. A formal product of prime-power residue spaces need not match the physical distribution, and arbitrary comparison functions could concentrate on rare congruence events. The proposed repair combined nilsequence stability under residue refinements with direct moment estimates. For two sampled points, the greatest common divisor of the coordinates of $t - t'$ controls their shared congruence condition. Small gcds suggested lattice-counting approximations; large gcds required a tail estimate. Approximate orthogonality and L^2 control were intended to avoid inflated pointwise bounds. For low-degree components, it considered global level estimates associated with Keller, Lifshitz, and Marcus [32], and checked their applicability while developing a separate mixed Hölder argument for dimension-independent bounds.

Attention then turned to the nilsequence stability needed by this transfer. Recalling its concern about the parameter dependence in Leng's multiparameter equidistribution approach [37], the assistant pursued an elementary reduction in degree based on lattice calculations. Transporting full factorizations across residue branches introduced conjugations and potentially new denominator primes. The proposed alternative tracked leading homogeneous symbols, whose behavior under translation was simpler. Independent rational constraints would shrink their common space; bounded degree and rank decreases were intended to control repeated reductions:

VERBATIM EXCERPT

Number of modes dimension poly. Iteration depth d bounded, so $\log S$ growth polynomial okay. This could solve! Let's formulate hierarchical nil regularity.

The assistant proposed canceling earlier scaling denominators after further residue refinement as the arithmetic mechanism for regularization. The assistant next considered smooth weights excluding wrapped cyclic progressions and an induction transferring domination by degree- d nilsequence tests to translated products tested at degree $d - 1$. The planned inverse-theorem input was the quasipolynomial result of Leng, Sah, and Sawhney [36]; their nilsequence recurrence estimate was to support the eventual return to progressions [35]. The assistant still needed to justify common-anchor alignment and lattice reconstruction. It reached the degree-one base as the final obstruction. Returning to the Kelley–Meka method [33], it reread Theorem 17 of Bloom and Sisask [4] to obtain regular Bohr sets of almost-periods for sifted convolutions, without completing the application.

19 From absolute increments to the relative lifting bottleneck

The sustained attempt to prove Erdős's reciprocal-sum conjecture now concentrated on transferring density increments into regions already defined by structured tests. The assistant continued developing the proposed absolute machinery. A shift-comparison argument used Bohr-set refinement and the

concave potential $(uv)^{1/4}$, where u, v were local means, to pay for extracting unusually dense pieces. High-moment unbalancing from Kelley–Meka and the local almost-periodicity theorem in Bloom–Sisask (Theorem 17) were intended to produce further refinements [33, 4]; a proposed nilsequence degree-reduction argument would extend the comparison beyond degree one. The assistant also used the Leng–Sah–Sawhney quasipolynomial inverse theorem to seek nilsequence correlations with polynomial logarithmic dimension bounds [36]. The proposed sampler remained unverified.

Positive-grid counting was then organized into packets of density increments. The relative step had to keep rank growth linear in the old and incoming dimensions, with suitable polynomial bounds on logarithmic complexity. Resetting the resulting structured region to an ordinary interval required a partition into arithmetic progressions, rather than isolated recurrence points. While checking the needed dimension dependence, the assistant recalled Green–Tao’s quadratic approximation argument [21] and considered Maynard’s simultaneous polynomial approximation bounds, questioning whether their constants would fit the iteration [43]. The assistant used the Leng–Sah–Sawhney progression-decomposition lemma, whose polynomial dimension dependence supported this reset [35]. The assistant calculated that sufficiently large packet gains, together with its transfer and reduction steps, would establish

$$r_k(N) \ll \frac{N}{(\log N)^3},$$

where $r_k(N)$ is the largest size of a subset of $\{1, \dots, N\}$ without a nontrivial k -term progression. Dyadic summation would then imply the reciprocal-sum conclusion.

The transfer itself exposed two obstructions: sampled nilsequence tests varied too freely to select a small fixed candidate family, and the sampler’s marginal density lacked the bound needed for a Hahn–Banach separator argument. The proposed repairs localized tests before discretizing their parameters and randomized constant polynomial labels to bound the marginal. The local-slice distribution still needed checking:

VERBATIM EXCERPT

| Yes! Need ensure t distribution for cap when local slices?

A weighted semidirect-group calculation subsequently sought the identity $F(b) = V(b) - V(b - x)$, integrating a closed polynomial one-form to obtain a controlled decomposition into slow and rational components. The assistant still needed cube comparison, scalar transfer, and uniform estimates for the conditioned sampler to complete the proposed route to the density bound and the reciprocal-sum conclusion.

20 Sampling geometry and the limits of the alignment framework

The attempt sought a density-increment proof that a set of positive integers with divergent reciprocal sum contains arbitrarily long arithmetic progressions. It proposed grouping local increments into packets and lifting each packet while increasing the number of floor coordinates only linearly. A reset invoking the Leng–Sah–Sawhney Schmidt decomposition [35] was then intended to produce a longer-scale progression with increased density. The assistant sought the bound $r_k(N) \ll N/(\log N)^3$ through this lifting step and used dyadic summation to connect it to reciprocal-sum convergence on progression-free sets.

The assistant also revisited a finite-field model: it hoped to regularize polynomial towers using Lampert–Ziegler’s relative-rank bounds [34], while explicitly leaving their required dependence on layer widths to be checked, and invoked Janzer’s polynomial comparison of partition rank and analytic rank [29] to transfer rank information from small restrictions. For the integer argument it used

the Leng–Sah–Sawhney quasipolynomial inverse theorem [36] to turn Gowers-norm detection into nilsequence correlation, and Bloom–Sisask’s local almost-periodicity theorem [4] for the degree-one shift comparison.

The main obstacle became a constrained affine sampler $\psi(t) = x + Vt$. Polynomial constraints were split into small continuous lifts and integer parts; rational subspaces introduced projected lattices whose heights could be enormous. Cancellation between continuous covolume and lattice spacing suggested a route to estimates independent of orientation, but the assistant had not resolved chart counting and normalization. Parameter order was equally consequential:

VERBATIM EXCERPT

We cannot choose m after post- b because b may be $(p+m)^D$ from all recursion. Need m fixed function p before defining b .

Here m is the sampling dimension, p structural complexity, and b a later accuracy budget. The developing construction classified lattice directions by scale, randomized polynomial coefficients, and used disjoint product blocks to seek Fourier decay. Cube comparisons would discard non-site Fourier modes through high rank and represent the remainder by separated site factors.

The assistant simplified the proposed density comparison to total variation control:

VERBATIM EXCERPT

Can avoid integration by parts many derivatives; just one L^1 variation. Yes. Need ensure cofactor inverse stable for actual and ideal on good support in v , boundaries smoothing.

A second simplification used two-point comparisons for scalar transfer, avoiding classification of arbitrary ambient linear frequencies. Positive weights combining real densities and congruence laws were meant to preserve density excess after removing large good primes.

The final scrutiny shifted to alignment: sample-dependent maps had to be replaced by one ambient polynomial map preserving a marked projection. Central quotients and fiber products offered an induction mechanism, but the assistant still needed to prove compatible slice refinements, uniform reconstruction, and quantitative thresholds to complete the sampler theorem.

21 From weighted lifting to contraction audits

The attempt on Erdős’s reciprocal-sum conjecture moved from constructing a relative lifting mechanism to testing its analytic foundations. The immediate objective was to convert structured tests that vary across affine samples into one ambient test while keeping the number of integer floor coordinates under control.

For a weighted homogeneous polynomial $D(u, b)$, the proposed decomposition separated a slowly varying polynomial from a bounded-denominator rational polynomial, but only on specified rational spaces K_h containing the existing high-rank directions. The assistant used the Leng–Sah–Sawhney inverse theorem [36] to turn local structured correlations into an ambient nilsequence correlation. A semidirect nilpotent group encoded the calculation. Bracket differentiation and a weighted Euler identity were intended to supply a rational potential. The assistant found that obtaining slow variation required substituting full constraint polynomials rather than just their leading terms.

Relative lifting then processed the floor coordinates by increasing weight. Tagged lower-degree copies were introduced so that the potential $\sum_h h \dim W_h$ decreased without uncontrolled growth of rational heights. The assistant proposed a final coordinate bound $d_0 + sd$, with recursive budgets fixed before the late rank and sampling thresholds. The proposed iteration reset to long progressions using the Schmidt

decomposition in Leng–Sah–Sawhney, Lemma 2.1 [35]. The assistant questioned whether this recursion preserved positivity.

The audit first examined scalar transfer: conditioning on prime-power residues, restarting after new denominator primes appeared, and controlling affine-sample correlations through moment estimates. It then concentrated on degree-one comparison over regular Bohr sets. Spike extraction was intended to pay for exceptional cells through the concave potential $(uv)^{1/4}$, where u, v are cell means. For the remaining cells, convolution moments and dependent random choice in the Kelley–Meka approach [33], together with Bloom–Sisask’s almost-periodicity theorem (Theorem 17) [4], were meant to produce variance and hence strict contraction.

The assistant found these checks locally favorable but continued to scrutinize extension of products to the higher-degree square construction after interval restrictions, together with quotient-kernel normalization. It continued checking the lifting and comparison arguments used to derive $r_k(N) \ll N/(\log N)^3$ and hence reciprocal-sum convergence of progression-free sets.

The assistant recorded the resulting sign of the contraction:

VERBATIM EXCERPT

| terminal zero. Thus γ positive.

22 Testing the proposed transfer architecture

The assistant now tests its proposed proof of Erdős’s reciprocal-sum conjecture. Its quantitative target is $r_k(N) \ll N/(\log N)^3$, where $r_k(N)$ is the largest size of a subset of $[N]$ without a nontrivial k -term arithmetic progression. Dyadic summation would then give the conjectured implication.

The assistant recalls the Kelley–Meka three-term bounds and their possible extension to higher-order boosting [33], alongside Green–Tao’s bound $r_4(N) \ll N/(\log N)^c$ [17]. It also recalls why the sparse graph counting approach of Filmus, Hatami, Hosseini and Kelman does not directly handle longer progressions: the binary-system argument requires an underlying simple graph [12].

The audit follows a dependency chain. First, bias in polynomial nilsequence tests should force a reduction of the structured path’s top-degree algebra. The assistant checks shift comparisons, lattice co-volumes, and removal of slow and rational coefficients against elementary nilpotent examples. Next, one-sided domination should propagate to lower-degree tests after multiplication by translated positive weights. A Bohr-set argument uses the concave potential $(uv)^{1/4}$; higher degrees require derivative correlations and positive reconstructions. The assistant also uses Bloom–Sisask Theorem 17 on local almost-periodicity [4] while developing these propagation arguments.

For the structural modeling step, the assistant checks and uses the quasipolynomial inverse theorem of Leng, Sah and Sawhney: at threshold e^{-p} , the nilsequence dimension is polynomial in p , while complexity and inverse correlation are exponential in a polynomial [36]. Positive counting estimates would yield an absolute density increment. The assistant finds that iteration still depends on a relative lifting statement whose rank grows linearly with the previous rank. Without that control, the length losses could overwhelm density amplification. To reset a block of increments on an unconditioned progression, it uses Lemma 2.1 of Leng–Sah–Sawhney’s *Improved Bounds for Szemerédi’s Theorem*, partitioning into progressions on which the niltest oscillates little, then discarding short pieces using the average-length bound [35].

Attention then narrows to scalar transfer along affine samples $t \mapsto x + Vt$. Prime-power conditioning is intended to stabilize a bounded niltest baseline. Degree descent and strict reductions of a fast subalgebra should terminate this process.

The assistant then reconstructs orthogonal residue-coordinate expansions, checks exceptional pairs

with large common divisors or small separation, and distinguishes complete coupling levels from incomplete levels requiring contraction. It recognizes that conditioning costs must be fixed before choosing the truncation cutoff. Revisions clarify lattice covers, covolumes, and separation arguments. It then returned to the transfer and relative lifting statements used in its iteration.

23 Parameter uniformity in the proposed density increment

The assistant continued auditing its own developing argument for Erdős's reciprocal-sum conjecture. The intended quantitative endpoint was

$$r_k(N) \ll N/(\log N)^3,$$

where $r_k(N)$ is the largest size of a set in $[1, N]$ without a nontrivial k -term arithmetic progression. Dyadic summation would then force every such progression-free set to have convergent reciprocal sum.

The assistant also revisited the Bohr-set almost-periodicity input from Bloom and Sisask [4] in its degree-one shift comparison.

The first audit sought to turn frequently observed local polynomial structure into one ambient patch. A weighted polynomial was to admit a slow-plus-rational decomposition. Encoding translations in a polynomial group produced a closed polynomial one-form, whose potential supplied the proposed integrability step. Marked projections, central averaging, and induction on degree were intended to preserve positivity and control the number of patch coordinates.

Attention then shifted to the sampling machinery on which that lifting depended. Cube comparisons were supposed to eliminate Fourier modes that failed to factor through the sampled vertices, using repeated Cauchy–Schwarz and a high-rank hypothesis. Product blocks supplied smoothing; Jacobian estimates supported total-variation control. The proposed passage from a large ambient Gowers norm to an ordinary nilsequence test used the Leng–Sah–Sawhney quasipolynomial inverse theorem [36]. The assistant emphasized comparing individual structured twists separately:

VERBATIM EXCERPT

| This comparison for **each** twist J no global joint test, so no lcm growth.

For scalar transfer, a positive forecasting weight combined continuous densities, inactive grid constraints, and conditional congruence laws. Removing good-prime factors required pointwise control. A concrete correction replaced a floor threshold by a ceiling threshold in the permanent sublevel argument.

The consolidated proposal combined these ingredients with an induction that lowered a weighted dimension potential whenever an old layer failed the rank test. Density amplification and decomposition onto long progressions were then meant to make density gains pay for length losses. For that decomposition, the assistant used Lemma 2.1 of Leng–Sah–Sawhney [35], seeking long progressions on which the nilsequence test varied negligibly compared with the score. But approximation bounds had to remain independent of late sampling scales, lattice heights, and prime-depth cutoffs.

24 Auditing transfer, lifting, and quantitative rank control

The assistant subjects its proposed route toward Erdős's reciprocal-sum conjecture to a sustained audit, moving from constrained sampling through scalar transfer to weighted lifting.

The intended density increment turns surplus density on a structured patch into another patch with controlled additional rank. Polynomial constraints take values in rational spaces W_h , with small real lifts

and integer carries. Disjoint product blocks, smoothing, and Fourier cancellation are intended to compare sampled cubes with ambient cubes while keeping early complexity bounds independent of later sampling scales and rational heights. Carry consistency and the regularity of the resulting densities receive particular scrutiny. The cube-to-ambient comparison uses Leng, Sah, and Sawhney’s quasipolynomial inverse theorem to turn a large ambient Gowers norm into nilsequence correlation [36]; the assistant also retains their effective Schmidt estimates among its stated inputs [35].

The next bottleneck is scalar transfer: positive scores on many local paths should yield a positive ambient score using only $\sum_h \dim W_h$ extra patch coordinates. A forecasting weight combines continuous, inactive discrete, and modular distributions. The assistant examines prime-power divisibility and Fourier tails to avoid paying for an enormous common modulus. The assistant continued seeking explicit estimates.

Weighted lifting then uses a rational potential and successive graded factorizations to align local patches. Positive localization selects standard small bumps, avoiding an excessive count of arbitrary Lipschitz functions. Rank preparation moves failed directions down one weight, with termination controlled by a decreasing potential:

VERBATIM EXCERPT

Potential many cuts $\ll (0(jD))$ because $(\sum_h \dim W_h)$ starts jD , each violation reduces dimension h by 1 and adds dimension $h-1$ line, net drop 1.

Here D counts the old slots at their lowest weight j . The assistant continues to check the proposed slot bound, simultaneous Fourier approximations, and product errors.

25 Quantitative repairs followed by a foundational audit

The assistant revises and audits its proposed proof of Erdős’s reciprocal-sum conjecture. Its target is

$$r_k(N) \ll N/(\log N)^3,$$

where $r_k(N)$ is the largest size of a subset of $[1, N]$ without a nontrivial k -term arithmetic progression. Dyadic summation would then force every such progression-free set to have convergent reciprocal sum.

The initial repairs concern quantitative dependence. Taylor arrays are compared with Haar measure through controlled Fourier or Lipschitz tests, without claiming total-variation convergence. Lattice-fiber estimates keep exponents dependent on a fixed number of output rows, rather than an expanding number of input coefficients. Slow and rational factor adjustments are elaborated before attention returns to the foundational step-drop and shift-comparison claims.

The audit checks how lattice covolumes, horizontal directions, and nested derivative brackets support the proposed reduction of nilsequence degree. For degree-one shift comparison, the key danger is rank growth: repeated increments must avoid dependence on the parent rank that would accumulate into a tower. Regular Bohr sets, density extraction, and almost-periodicity are examined with this requirement in view.

The assistant checks the hypotheses of Bloom–Sisask’s local almost-periodicity theorem, especially its rank increment, while auditing degree-one shift comparison [4]. It also checks the progression-partition lemma of Leng–Sah–Sawhney: the proposed iteration uses nearly constant nilsequences on long progressions to reset a structured density increment [35]. Its nilsequence detection and modeling arguments use their quasipolynomial inverse theorem for the Gowers norm [36].

The subsequent synthesis organizes positive structured patches, scalar transfer, constrained sampling, and lifting into a proposed density-increment architecture. It targets rank linear in existing and newly

introduced patch slots; packaged density gains are intended to pay for progression-length losses. The assistant also requires higher-degree reconstruction to preserve positivity. The assistant ends by formulating detailed claims linking transfer, sampling, lifting, and the inverse theorem within this proposed architecture.

26 Quantitative audits and a renewed circularity concern

The assistant now audits whether its proposed proof of Erdős's reciprocal-sum conjecture can support all its quantitative dependencies, including its use of Bloom–Sisask almost-periodicity (Theorem 17) [4]. It rechecks the quasipolynomial inverse theorem of Leng, Sah, and Sawhney for the bounds on nilsequence dimension, complexity, and correlation [36], and their Schmidt decomposition lemma for the progression reset and its length and oscillation estimates [35].

First, scalar transfer must preserve correlation with a nonnegative structured nilsequence under affine sampling. Residue conditioning and an orthogonal decomposition by coordinate subsets are intended to control exceptional slices. A normalization issue appears: the sampling-density domination may exceed the error budget unless the logarithmic parameter is enlarged first. The assistant proposes that repair while also checking a novel step-drop lemma, positive progression counting, and whether repeated refinements leave enough interval length for a final density contradiction.

The audit then reconstructs the polynomial sampler, separating lattice coordinates into inactive, moderate, and enormous scales, alongside continuous directions. Dedicated products of variables generate polynomial terms; differencing and rational approximation are supposed to suppress unwanted Fourier modes.

The assistant identified cube-density comparison as the next difficulty: its regularity bounds deteriorated as perturbations shrank, so it could not use them to dismiss those perturbations. The proposed alternative compares the ideal and perturbed pushforward laws using Jacobian bounds and total variation. Further checks concern modular forecast weights, parameter ordering, and nilpotent lifting without inflating the rank of the resulting structured patch.

The assistant calculated that its assembled iteration would give

$$r_k(N) \ll N/(\log N)^3,$$

where $r_k(N)$ is the largest size of a subset of $[N]$ without a nonconstant k -term progression. Dyadic summation would then force every such set's reciprocal sum to converge.

A remaining concern is that a physical mesh depends on a truncated expansion whose bound already depends exponentially on the truncation parameter. The assistant leaves this circularity concern unresolved.

27 Closing scrutiny of the proposed reciprocal-sum proof

The attempt ended by auditing its candidate proof of Erdős's reciprocal-sum conjecture. Its target remained

$$r_k(N) \ll_k N(\log N)^{-3},$$

where $r_k(N)$ is the largest size of a subset of an interval of length N without a nonconstant k -term arithmetic progression. Dyadic summation would then force every such progression-free set to have convergent reciprocal sum.

Closing checks examined whether shifted polynomial nilsequence paths retained the filtration needed for degree reduction after normalization and quotienting. They revisited conditional moment inequali-

ties, Chinese remainder theorem comparisons, rational lattices, constrained affine sampling, and potentially circular parameter choices. A correction used pigeonholing to obtain a common cube slice from short-axis choices.

Batching density increments risked excessive rank growth. With p_0 a logarithmic inverse-density scale and G a fixed density-gain factor, the proposed packet-rank bound was $C_G(2 + p_0)^{D_s}$, with exponent independent of G . The assistant calculated that repeating $O(\log p_0)$ packets would preserve polynomial rank, letting density gain exceed the loss when nilsequence decomposition reset the patch to an interval progression. The assistant checked for hidden dependence that could defeat this calculation.

The assistant rechecked the Leng–Sah–Sawhney inverse theorem’s dimension and Lipschitz bounds for its nilsequence correlations [36], and the progression decomposition in their quantitative Szemerédi paper for resetting patches [35]. It also examined the relative-density and regularity hypotheses of Bloom–Sisask Bohr almost-periodicity in the degree-one comparison [4]. Its final response presented the proposed argument.

Part II: The quasipolynomial strengthening

Original prompt (excerpts)

Problem

Please give quasipolynomial bounds for k -th arithmetic progressions for all $k \geq 3$. Use the previous bounds you have done.

28 The stronger target and the cost of precision

The assistant seeks $r_k(N) \leq C_k N \exp(-c_k(\log N)^{c_k})$, equivalently a threshold $N \geq \exp(p^{C_k})$ at density e^{-p} . Starting from the earlier argument, it isolates a quantitative obstruction: relative lifting adds patch ranks, but repeatedly raises logarithmic resource bounds to a fixed power. The recurrences

$$H_{j+1} = \kappa H_j^2, \quad P_{j+1} = P_j + (2 + P_j)^E$$

amplify density rapidly while still giving $\log P_j \ll E^j \log(2 + p)$.

Leng–Sah–Sawhney’s inverse theorem [36] supplies nilsequence structure, and their progression partition [35] resets a positive patch score to an ordinary density increment. Schoen–Sisask’s radius-sensitive almost-periodicity, Theorem 5.4 [49], and Bloom–Sisask’s Theorem 17 [4] support the degree-one comparison; Bohr widths must be tracked separately from rank.

The assistant seeks constant-factor growth of logarithmic resources through $O(\log p)$ levels; a factor p^C per level would already exceed its budget. Structured positive models offer another route, but smaller counting errors could enlarge their complexity. It returns to positive grids and bilinear kernels.

The assistant isolated the difference between additive rank growth and repeated precision loss. The earlier lifting argument returned at most $d + d_0$ slots from d old slots, but could replace a logarithmic budget P by $(2 + P)^E$. Preparing and freezing successive weights controlled one lifting operation; iterating whole operations still threatened the desired threshold $\exp((\log(1/\alpha))^{O_k(1)})$.

It explored local inverse theorems normalized by patch mass, tensor amplification, reductions to shorter progressions, and multidimensional affine boxes. Tensor products multiplied the density loss,

while carry avoidance spoiled the apparent economy of sphere constructions. Kelley–Meka’s large-moment method [33] inspired polynomial-size grid ideas, but binary cubes did not themselves contain the needed progression. Retaining box volume looked promising for linear Bohr sets; quadratic mixed-coefficient constraints appeared to consume that advantage.

Encoding precision in unrestricted coefficients introduces unwanted integer branches. Auxiliary slots do not exclude them automatically. Height-independent lattice charts suggest treating continuous and discrete directions separately, while the assistant continues seeking density gains without repeatedly powering the old precision budget.

The assistant seeks to separate the precision needed inside scalar transfer from the complexity of its output patch. In the earlier argument, slot counts remain additive, but residue slicing, lattice heights, and torus-cell selection feed old precision into new widths and scores. The desired replacement would charge those costs to side lengths while shrinking old residual coordinates only by $\exp(-p^{O(1)})$.

It explores normalized polynomial tubes, generic-path independence, and degree-stratified losses. The quantitative benchmarks are Szemerédi’s theorem and Gowers’s proof [50, 13], together with Green–Tao, Kelley–Meka, and Leng–Sah–Sawhney [17, 33, 35]. Entropy, high moments, and simultaneous increments suggest possible alternatives to repeatedly conditioning on tiny cells, but no suitable global model is obtained. Seminorm modelling with the Leng–Sah–Sawhney inverse theorem [36] and cube lifting supplied by OpenAI in *Monochromatic finite sums and products in the positive integers* [46], in the Host–Kra framework [27], motivate further transfer ideas.

Tensor powers dilute gains; translated intersections fail to align differences; latent coordinates introduce vertical progressions. The assistant seeks large internal dependence on old precision Q with output-width costs polynomial in p .

VERBATIM EXCERPT

Height encoded but pre-precision governs occupancy; to target a rare high-density cell cannot ignore tiny mass for detection/global inverse. Could use full old complexity in proof of existence/globalization without incurring it in output patch Lip!

VERBATIM EXCERPT

Need domains retain structure for each of $O(p)$ increments at cost $\exp(-p^C)$ factor independent of old precision.

29 Normalized symbols and relative lifting

The assistant sought a two-budget strengthening of the earlier lifting mechanism: new density gains should cost $\text{poly}(p)$ even when old chart widths are e^{-Q} , with $Q \gg p$. Substitutions, sampling, and entropy models threatened degree, scale, and intersection control. The reference framework retained Leng–Sah–Sawhney’s inverse and partition results [36, 35], Schoen–Sisask almost-periodicity [49], Bloom–Sisask’s three-term comparison [4], and Kelley–Meka’s multiplicative increment perspective [33].

A possible separation appeared when a degree- h chart was tested by degree- r functions with $r < h$. The assistant proposed using independence of polynomial tensors on cube grids to transfer normalized correlations. A signed dense model with cap $e^{\text{poly}(p)}$, obtained through quadratic rather than high-product comparisons, might avoid the costs of a model bounded by one; Conlon–Fox–Zhao’s relative-removal framework [7] was a point of comparison. For $r \geq h$, dependencies among constrained cube vertices blocked the same argument.

The assistant then seeks a shadow domain where the narrow chart has bounded width. Pulling increments back would require matching slow, rational, and fast statistics while preserving integrality, boundedness, progression counts, and joint laws.

The assistant sought to widen a tube $|C_h(u) - \beta| \leq e^{-Q}$ while preserving its normalized polynomial statistics. Such a shadow might make the earlier relative-lifting framework usable with losses controlled by density rather than accumulated precision. Doubling chains connected local identities to integral endpoints. Altering denominators raised congruence and product constraints.

Marginal face-symmetry lifts supplied by OpenAI in *Monochromatic finite sums and products in the positive integers* [46] suggested randomized joint shears, still obstructed by rational periods.

It then tried to keep internal algebraic matching at fine precision while returning a coarser test. Large rational slopes could restore precision feedback. Leng–Sah–Sawhney inverse-theorem models [36] did not automatically avoid that loss.

Direct counting on highest-degree factors borrowed positive comparison from Kelley–Meka [33]. The assistant found that variation in lower coordinates could encode four-term coloring problems. Positive models faced clipping costs.

The assistant targets $O(p)$ fixed increments with additive width losses and linear rank growth. Green–Tao’s local U^3 inverse method [18] and four-term bound [17] suggested a derivative-based route, but integrating those derivatives through sparse tubes remained unresolved.

VERBATIM EXCERPT

Violation cannot be distributed; amplification pointed from first slow requirement to endpoint. So simultaneous realization fails even approximate.

The assistant tried to preserve a useful density gain while confining old precision costs to a separate cold budget. Here the warm budget tracks the less expensive width losses, while the cold budget tracks more expensive algebraic costs. Instead of freezing a local test on tiny slices, it proposed a hybrid *EPR*: a slow real factor, an ambient polynomial map, and a rational residue factor. The propagation architecture from the previous argument, the Leng–Sah–Sawhney inverse theorem [36], and Leng’s efficient equidistribution theorem [37] supplied the contemplated detector and symbol-reduction tools.

Factoring the ambient map suggests replacing its fast state by an independent Haar variable. Simultaneously treating old constraints might limit added coordinates to the new local group. The assistant still needs globalization preserving weighted degree and positive score.

Composition can increase weighted degree, while residue representatives may require one new coordinate per old variable. The assistant seeks periodic outputs, then considers degree s^2 and averaging away coordinates above s . Positivity and the rank cost of high-weight residue slots remain obstacles.

VERBATIM EXCERPT

Real mesh of u can be in slice cost only; width of y_h determined by chart-relative smooth derivatives $e^{\{\text{poly}(p)\}}$ if t -slice simple.

30 Affine restrictions and degree control

The assistant allows ambient degree $D \approx s^2$ with detector degree $s = k - 2$. Positive conditional averaging would remove high-weight coordinates. Affine-plane inverses, Hensel lifting, and the Chinese remainder theorem might preserve the test group without adding residue slots for every old coordinate.

The unresolved issue was simultaneous preparation. Processing old layers separately could yield $s^{2^D} > D$, defeating a fixed degree bound. The assistant returned to the sampler, fixed-group globalization, and descending rank potential of the previous argument, seeking losses independent of old widths.

The recalled benchmarks were Gowers, Green–Tao, Bloom–Sisask, and Kelley–Meka [13, 17, 3, 33, 5]; inverse and partition inputs were Leng–Sah–Sawhney’s general and five-term work [36, 35, 39]. Nilsequence and positive-comparison foundations were attributed to Green–Tao, Green–Tao–Ziegler, Schoen–Sisask, and Bloom–Sisask [23, 22, 49, 4]. Historical density, prime, and coloring results were attributed to Erdős–Turán, Szemerédi, Green–Tao, and van der Waerden [10, 11, 50, 20, 51].

The assistant sought to replace repeated polynomial resource growth by a hierarchy of width budgets. Reviewing the machinery of the previous argument, it contrasted the manageable rank recurrence $R_j = 2^j p^{O(1)}$ with $P_{j+1} = P_j + (2 + P_j)^E$. The reset based on Leng–Sah–Sawhney’s nilsequence decomposition [35] made dimension especially important: its progression-length exponent deteriorated polynomially with dimension.

Finite-field and box models lose density when preventing wraparound. The assistant instead processes constraints by increasing weight, retaining top-weight coordinates and charging precision to lower determining coordinates. Added coordinates would flow downward through finitely many weights, but their total rank still needs control.

Tiny higher-layer support can concentrate scores on rare paths, defeating normalized-density estimates. Replacing $E(\theta)hD(\theta)$ by weighted polynomial factors also needs uniformity in unbounded h . Averaging additional coordinates suggests a route, with lattice covariance and dimension still unresolved.

The assistant focused on preserving rank while replacing local tests by ambient ones. The earlier triangular-patch conversion separates unrestricted coefficients from kernel precision and rank. Recursive amplification seemed affordable in rank, but repeated path-length losses could turn a fixed polynomial budget into $p^{C \log p}$. Constant increments therefore required additive or strictly hierarchical rank growth.

For a hybrid test $F(E(\theta)P(u, \beta)R(\theta))$, the desired replacement would eliminate θ without increasing weighted degree or introducing slots for bounded representatives of the fast variable. Large lattice shifts and small continuous smoothing were explored, but transferring the resulting arbitrary test required conditional distribution estimates with potentially circular precision costs. Symbol reduction used the framework of Leng’s efficient equidistribution theorem [37] and the Leng–Sah–Sawhney inverse theorem [36].

Reduction modulo $B(\theta) - X$ suggests comparing exact fibers with coarse neighborhoods uniformly in the fast variable. Products recreate eliminated components, and subpower Fourier decay does not give summable tails. The assistant therefore revisits relative strength and independent product-block estimates.

VERBATIM EXCERPT

At depth $\log p$ yields $p^{C \log p}$. Need avoid using amplified rule recursively (side length multiplication). Thus constant increment iteration only: rank growth must be additive/hierarchical, widths same.

VERBATIM EXCERPT

Could use Hensel linearization of polynomial map along some structured affine subspace of t to right-inverse low-coordinate variables as genuine linear polynomials with rational coefficients mod M plus small-modulus restrictions.

31 Separating internal precision from output width

The assistant compares exact fibers $B(\theta) = X$ with coarse neighborhoods under tests $F(g(\theta, h)\Gamma)$ uniform in h . Coupling between blocks defeats independence. Fiber ideals, oscillatory estimates, and affine sampling face recreated product terms and integer denominators.

Using the symbol and globalization framework of the previous argument, it next considered a smoothed model $E(X)J(h)R(X)$. A candidate depending on latent h still had to become a polynomial expression in the actual variables. Preserving that projection became a central requirement.

The surrounding quantitative benchmarks were Szemerédi [50], Gowers [13], Green–Tao [17], Kelley–Meka [33], and Leng–Sah–Sawhney [35]. The assistant sought to avoid repeated losses by retaining passive old constraints and transforming only active low-degree data. A finite palette with J layers might replace repeated multiplication by layer weights with $r \mapsto O(r^2)$, giving a cap s^{2^J} . The final idea preserved a dependency graph between levels. Rank-preparation downgrades, integer carries, and short parameter directions remained obstacles to closing that scheme.

VERBATIM EXCERPT

Cannot tensor independence B blocks with arbitrary nf test coupling all blocks. But can Cauchy-Schwarz and apply high-rank inverse estimate with dimensions m huge; exponent degrades $\text{poly}(m)$, rank maybe enough but circular.

Separating active variables from passive masks still permits $R_i \simeq R_{i+1}^2$. The assistant proposes simultaneous preparation by effective degree, charging rank failures to lower-layer widths.

Cube comparison, detection, and strong-projection refiltration from the reciprocal-sum argument supplied the starting mechanisms. The assistant also reconsidered Conlon–Fox–Zhao relative transference [7], Kelley–Meka boosting [33], and shift comparison with its Schoen–Sisask almost-periodicity input [49]; these did not resolve the dependence on old chart precision.

Intermediate integer directions are too expensive to freeze and too short for approximation. Linear sampling recovers parameters but leaves path-dependent translations. Fixing them introduces derivative constraints; polynomial families instead encounter refiltration across different sites. The assistant returns to fixed translations.

Expanding $\beta_i(x + Vt) = \sum_{|\alpha| \leq i} b_{i,\alpha}(x)t^\alpha$ lowers coefficient weights to $i - |\alpha|$. The assistant seeks downward dimension flow, but conditional rank and density need control. Simultaneous sampling may again charge narrow lower widths to higher layers.

The main alternative was algebraic elimination. For a smoothed model $P^* = \Phi(X, h)$, a replacement depending on latent h would not automatically transfer to the original variables. The assistant sought a multiplicative substitution L satisfying $L(B(t)) = X$, preserving slow-factor bounds, rational denominators, and nilpotent reconstruction identities. The assistant considered Ananyan–Hochster, Stillman, and strength ideas as possible approaches to this substitution [1].

Detection continued to use Leng–Sah–Sawhney’s quasipolynomial inverse theorem [36]; its dimension dependence created a potential circularity between partner complexity and sampler strength. The earlier lifting framework supplied the structural starting point. Simultaneously respecting real smallness and rational integrality remained the central difficulty. The assistant continues comparing these elimination routes and their cumulative costs.

VERBATIM EXCERPT

| Need handle macro elimination on integer fibers.

32 The problem of repeatedly returning to intervals

The assistant seeks relative comparison independent of scale: additive rank alone cannot prevent repeated precision losses from defeating $N \geq \exp(p^{O(1)})$.

It revisited the counting and lifting mechanism of the earlier reciprocal-sum argument. The recurrence $H_{j+1} = \kappa H_j^2$ amplifies gain, but $P_{j+1} = P_j + (2 + P_j)^E$ compounds resources. Proposed alternatives included direct comparison on normalized thin patches, a hierarchy of residual widths, averaging enriched cells, and digit-box embeddings. Rare-cell mass losses, spherical layers, and integer carries frustrated these routes; no scale-independent replacement was completed. Each proposal still needed uniform control before later sampling parameters were chosen.

The reviewed historical context comprised Erdős–Turán, Szemerédi, Erdős’s reciprocal-sum question, Green–Tao on primes, Green’s survey, and van der Waerden [10, 50, 11, 20, 25, 51]. Quantitative benchmarks were Gowers, Green–Tao, Bloom–Sisask, Kelley–Meka, and Leng–Sah–Sawhney [13, 17, 3, 33, 4, 5, 39, 35]. Nilmanifold equidistribution, inverse theorems, and almost-periodicity supplied the structural background [23, 22, 49, 36]. The assistant’s closing direction was to seek a larger density gain at fixed degree without repeatedly conditioning on exceptionally small cells.

The assistant revisited the earlier reciprocal-sum argument’s sampling, lattice-chart, and lifting foundations. Rank preparation lowers the potential $\sum_h h \dim W_h$, but this dimension accounting alone does not control the precision costs of successive density increments. Inactive integer projection coordinates were the main obstruction: their bounded ranges prevent polynomial variation along arbitrarily long paths.

Fixing a lattice sheet simplifies identities but makes thresholds depend on covolume. Positive character tests cannot isolate it cheaply. Keeping sheet variables requires replacement tests to use partner information from one ambient site; dummy variables and shadow models do not resolve this.

Full shear groups could repeatedly replace rank by a polynomial. The assistant separates orientation from approximation: relation rows remain bounded at original resolution, while denominators depend on conditional mass. Sampling frequencies may still enlarge those rows, motivating separate ensembles or scales.

VERBATIM EXCERPT

| We need decisive simplification. Let's attempt direct path selection global via *exact polynomial maps on high-rank varieties* and L map applied to cold globalization without macro separate.

Short integer directions can remain family parameters, but differentiation risks rank feedback. Productive paths may carry only a tiny fraction of mass. The assistant seeks prefix-by-prefix mass comparisons that survive globalization.

Kelley–Meka’s degree-one method [33] motivated another examination of concentration removal, concave potentials, and matched local scales. The earlier reciprocal-sum argument’s relative-lifting and refiltration mechanisms offered additive rank, but still allowed old widths to drive subsequent complexity.

For short-parameter families, the proposed repair used evaluation pivots and filtration slack. Its unresolved point was reconstructing all marked functions at one common spatial site when pivot relations

involved different sites. Freezing sheets was also costly: for $C = (P, BP)$, fixing $b' - Bb$ can impose a width of order $1/B$ on the fractional part of P .

Enlarged domains, finite-field models, and surrogate laws leave a recurring obstacle: tiny structured correlations do not force alignment on short paths.

VERBATIM EXCERPT

Old slot complexity maybe rank $O(\text{poly}(p))$ widths tiny $\exp(-\text{poly}(p))$; losses in outputs depending old widths lead runaway recursion.

33 Constrained sampling and scalar transfer

The assistant rejects an ultraproduct shortcut because p^{CEj} does not give a uniform exponent. A factorization stable in conditional total variation would tolerate oscillatory tests, but higher-degree constraints and tiny residual coefficients obstruct polynomial transversality.

Using the step-drop framework of the earlier reciprocal-sum argument, with antecedents in Leng's efficient equidistribution theorem [37] and Leng–Sah–Sawhney's degree reduction [36], it proposed splitting integer outputs into long directions and short coordinates z . Extracting the least short-weight term $z^\alpha D_\alpha$ was intended to sharpen slow bounds without allowing narrow earlier widths to enlarge rational denominators. Rank failures would then move downward in degree.

A test variable ζ and family variable z' permit a marked-data correction that vanishes on the diagonal and preserves the leading long-variable symbol. The assistant seeks recursive degree reduction with diagonal reconstruction, while checking degree caps, constant terms, and productive mass.

VERBATIM EXCERPT

To show there exists $c > 0$ all N , need uniform exponent, cannot overspill with varied E .

The assistant developed an affine-section approach within the earlier reciprocal-sum argument's sampling and modular-forecasting framework. For inactive coordinates it proposed retaining the degree- h product blocks while adding a free linear variable. Fixing the product variables would then make exact inversion possible without sacrificing the leading-degree structure.

Active outputs need many affine sections with controlled incidence. Exceptional probabilities can swamp tiny scores, while increasing dimension risks circularity. Independent coefficients on disjoint columns, second moments, and nonsingular Hensel lifting suggest weights bounded everywhere and near one outside exceptional residues.

The assistant tests $f\mathbf{1}_{\text{good}}$, using mixed centers to protect mass and truncated inclusion–exclusion to replace primewise masks by bounded-modulus tests. Degree relaxation threatens passive-layer separation, prompting another examination of warm detection and signed models.

The assistant sought to prevent narrow old charts from making each density increment polynomially more expensive than its predecessor. Starting from the earlier reciprocal-sum lifting mechanism, it proposed a buffered invariant

$$\mathbb{E}fB_{\text{strict}} \geq a \mathbb{E}B_{\text{buffer}}.$$

This would protect positive mass from tangencies to chart boundaries. Passive determining masks would remain ambient functions; active ones would be preserved exactly by a marked projection while new coordinates alone acquired larger degree. The recalled progression-reset comparison with Leng–

Sah–Sawhney [35] motivated this stricter accounting.

A signed dense-model approach then encountered a circularity: improving approximation enlarged the model cap, and the Leng–Sah–Sawhney inverse theorem [36] could return a correlation smaller than the approximation error. An L^2 model merely shifted the difficulty to truncation.

Drawing on Conlon–Fox–Zhao densification [7], the assistant proposed replacing non-root cube factors sequentially by bounded signs of conditional dual functions. Weighted grid estimates would preserve correlation; after only the root remained sparse, bounded duals could be replaced successively by niltests using the ordinary inverse theorem. This was intended to detect the root without simultaneously matching a large test family.

A further proposed free linear parameter would make inactive integer outputs invertible after freezing nonlinear parameters. The assistant continued checking rank preparation, buffered normalization, boundary estimates, and the quantitative savings.

VERBATIM EXCERPT

Absence of AP is global strong deficit, requires correlation affecting almost all f or severe concentration.

34 Rank preparation and layered width budgets

The assistant reopened the earlier reciprocal-sum argument’s sampler and globalization machinery. For constraints $C_h - c_h = y_h + \beta_h$, detection bounds had to be fixed before the sampling length, rank threshold, and subspace heights entered. The rereading emphasized height-independent lattice charts, the distinction between abstract coefficient normalization and normalization of the actual tilted law, and cube-image errors controlled by total mass rather than a large density supremum.

Globalization selected polynomially many significant top-frequency pivots, approximated masked inputs uniformly over path-dependent tests, and reconstructed exact marked projections through lower-degree induction. Yet amplification still paid

$$H_{j+1} = \kappa H_j^2, \quad P_{j+1} = P_j + (2 + P_j)^E,$$

so repeated polynomial resource composition limited the useful depth. The reset relied on Leng–Sah–Sawhney’s *Improved Bounds for Szemerédi’s Theorem*, Lemma 2.1 [35], to convert positive nilsequence score into a long-progression density increment.

The assistant therefore sought a recurrence linear in old resources, or a normalized weighted-major decomposition avoiding repeated powers of thin chart widths. A recalled macro-transfer route also considered the Leng–Sah–Sawhney inverse theorem [36], with weighted-degree and discretization difficulties. Revisited candidates included preparing constraints globally by effective degree, treating high tags as passive, allowing bounded degree overshoot, and substituting low tags through affine fibers. The assistant identifies simultaneous preparation, residue periodicity, normalized transfer, and repeated rank growth as obstacles to these candidates.

Inactive directions can be too short for symbol matching yet too costly to freeze. Polynomial families need common reconstruction, and lattice sheets risk circular dependence between rank thresholds and covolumes.

It revisited the rank-preparation and constrained-sampling mechanisms of the earlier reciprocal-sum argument. Moving a removed direction from tag h to $h-1$ decreases $\sum_h h \dim W_h$, bounding the number

of failures. Solving all relation rows at a tag jointly was intended to avoid repeated height exponentiation. Extending this bookkeeping to changing lower-layer charts remained unresolved.

Strict and buffered boxes satisfy $\mathbb{E}f1_{\text{strict}} \geq a\mathbb{E}1_{\text{buffer}}$. Their gap protects against tangency during refinements. Mixing feasible centers should normalize chart mass and retain productive paths, while non-site Fourier modes and torus covers still require treatment.

VERBATIM EXCERPT

Instead of replacing B with selecting chart cells immediately, insert new slots with unrestricted centers and update evaluation map candidate masks. Need measure concentrations only after prep complete.

The assistant investigated whether global rank preparation could preserve relative patch widths without charging narrow lower layers to higher ones. Using the earlier reciprocal-sum argument's triangular-patch, polynomial-image, modular-exception, and pathwise-detection machinery, it sought a failed-rank decomposition into a slow polynomial, a rational polynomial with controlled denominator, and a lower-degree remainder.

The assistant separates real bounds in normalized residual coordinates from rational denominators independent of lower widths. Formal-slot descent suggests subtracting slow and rational terms separately. Short axes and spatial rescaling motivate degree induction and modular nondegeneracy before pigeonholing.

Translated boxes preserve fixed strict/buffer margins. Rank removal moves directions downward while retaining exact higher substitutions. A two-variable family correction preserves the common-site projection on the diagonal; the assistant checks compatibility when additional axes become short.

VERBATIM EXCERPT

Maybe choose W full always (avoid inactive) by keeping lowered directions as independent coordinates but altering polynomial to generic high-rank perturbations?

35 A triangular density-increment proposal

The assistant refined a layered density-increment scheme using the earlier reciprocal-sum foundations. It found that restoring a polynomial map through a quotient section could destroy the required regularity. Its proposed repair postponed all kernel-translation averages until after scalar transfer and affine-plane inversion, imposing regularity only on the averaged bottom-level test.

The assistant retains modular good flags to protect tiny scores, using truncated inclusion–exclusion over small prime products. Free linear parameters invert inactive integer coordinates; affine sections and real smoothing address normalization and boundaries.

For passive layers, the assistant proposed densifying a sparse cube average one vertex at a time before applying Leng, Sah and Sawhney's quasipolynomial inverse theorem [36] to bounded dual functions. This avoided directly inverting a function with the large cap imposed by a narrow chart. The resulting strict/buffered patch certificate was $\mathbb{E}fB^- > a\mathbb{E}B^+$. The intended bookkeeping let dimensions grow from higher blocks and each logarithmic width loss depend only on higher widths. The assistant calculated that, with those preparation and transfer bounds, repeated constant multiplicative increments would yield the desired quasipolynomial progression bound.

VERBATIM EXCERPT

| Cannot ensure at least many V for all t if B level set singular at some t . Flags needed.

The assistant tried to keep rationality costs from becoming width losses. Using the earlier reciprocal-sum machinery, it separated bounded real inverse substitution from denominator control by exceptional prime powers. The recalled detector used Leng–Sah–Sawhney’s quasipolynomial inverse theorem [36] for ambient nilsequence partners.

Sharpening slow-plus-rational splitting precedes expansion in excluded coordinates. Degree induction should cancel side losses without iterating once per coordinate. Tensor terms require separate exact agreement; constants use pigeonholing. Rank failures move downward, and translated boxes preserve surplus.

Affine sections satisfy $b(x + V(s, h), r_0 + h) = b(x, r_0) + s$. Plane-invariant selection and nonsingular lifts should give near-one weights outside rare flags. Forecasting truncates flag inclusion–exclusion by small witness moduli to retain terminal paths.

Backward lifting keeps passive masks as site functions and active projections exact. Actual and dummy short variables support corrections. Representative selection waits until averaging supplies smoothness and periodicity; bounded active degree inflation should close the hierarchy.

The assistant separated its forward construction from the backward globalization still to be supplied. Using the earlier reciprocal-sum foundations, it sought to iterate certificates $\mathbb{E}fB^- > a\mathbb{E}B^+$ for strict and buffered polynomial patches. The crucial requirement was that the loss in $Q_i = \log(2/w_i)$ depend on higher widths, not on Q_i itself; descending induction would then control $O(\log(1/a))$ increments.

Normalized residuals control slow coefficients separately from exceptional denominator factors. Failed relations move lower-degree remainders downward while preserving exact higher substitutions. Shifted coverings are intended to protect surplus without inverse-mass losses.

Scalar transfer retains inactive parameters while duplicating parallel ones into real and residue variables. Affine planes and nonsingular lifts control modular flags through bounded witness moduli. The assistant obtains a forward fraction ca of productive terminal paths and turns to restoring their gain backwards.

VERBATIM EXCERPT

| Any mid axes v newly short after choosing later threshold; restricted symbols pure-long independent their values because g has $\deg \leq \text{layer}$ in v , so discarded v only lowers.

36 Globalizing the active and passive layers

The assistant inspected the earlier reciprocal-sum machinery, concentrating on exact projections and costs that must remain independent of later sampling scales. Globalization selects a basis of significant top frequencies, attaches ambient partners through uniform masked-input approximation, and intersects rational fast algebras. Injectivity at the top permits reconstruction through

$$H \simeq (H/H_r) \times_{H_F/H_{F,r}} H_F.$$

Slow and rational factors require corrections in the actual kernel, followed by compatible lattice refinements and controlled nets of the resulting test families.

Polynomial translation groups connect symbolic factorization to slow-plus-rational decomposition. The sampler tracks coefficient normalization, spatial equidistribution, and residues. Scalar transfer discounts good-prime density factors and postpones orientation costs until recovery.

The assistant keeps passive masks external and active slots exactly marked. It must preserve higher widths through candidate replacement, congruences, and degree changes. Same-weight dependencies wait until final descent; it next checks these inherited estimates.

VERBATIM EXCERPT

```
Major issue replacing candidates (globalization) without consuming w_j. For active j layers r maybe
high, detection cold in Q_j; symbolic lifting uses P_j coefficients unconstrained; want paired
score width lower at j = w_j exp(-poly(warm)), at all >j same.
```

The assistant audited the earlier reciprocal-sum machinery, including the use of Leng–Sah–Sawhney’s quasipolynomial inverse theorem [36]. It checked cube-count normalization, path-dependent detection, weighted densification, and clipping of separating functionals. The guiding concern was whether narrow current-layer widths could contaminate bounds meant to depend only on higher-weight data.

The modular audit permits bounded products of witness prime powers, avoiding one oversized modulus. It checks coefficient independence, real/grid perturbations, sublevel depths, primitive-lattice denominators, and preservation of additive rank $d + d_0$.

For a passive layer $j > s$, degree prevented the new degree- s symbol from involving degree- j lift variables. The proposed globalization first obtained a decomposition with width-dependent bounds, then sharpened it through long spatial columns and sequential modular filters. To preserve signed scores, it tracked both nonnegative components of

$$(fL^-)\Phi^- - \lambda L^+\Phi^+,$$

using finite nets of site-dependent fields. The remaining passive-transfer plan removed good-prime density factors through $1 + O(p^{-2})$ estimates while retaining exceptional primes and stride divisors. The assistant then turns from these parameter choices to the ascent.

Inactive columns and continuous pivots reduce reconstruction to constraints in $X = (u, \beta)$. Boundary strips are removed only from positive terms; short integer ranges retain exact membership. Smooth-cell averaging should then produce a parent certificate.

For active layers, selected affine planes give $x = a + V(B, r)$. The assistant argues that bounded total plane measure permits selection without a loss depending on the huge modulus. Exact periodicity under changing residue representatives is essential. The nilpotent-degree induction uses polynomial families in short variables, significant vertical Fourier modes, and central averaging. Its symbol-splitting and reconstruction inputs come from the earlier reciprocal-sum argument.

Correlation subslices identify symbols without altering the objective slice. The assistant separates final thresholds from later induction and unions over individual approximants to avoid excessive complexity. It checks projections and periodicity recursively, while still needing to preserve the widths on already processed layers.

37 Iterating with separate degree budgets

The assistant proposed a downward iteration preserving a signed, full-box-mass score, drawing on the earlier reciprocal-sum argument. Paired smooth cutoffs allowed small target discounts to absorb coverage losses, while old determining equations remained exact.

Passive layers use a common ambient orbit and retained modulus. Active layers use stochastic replacements with short-variable polynomial labels, common-input correction, refiltration, and fiber reconstruction. Only expected evaluations must be independent of residue representatives.

Extraction solves inactive coordinates, removes determinant and boundary exceptions, and uses affine residue sections. Averaging fixes constants while preserving surplus; spatial cells restore two cutoffs with inflation confined to absolute slots.

The assistant claimed a recurrence in which the logarithmic width cost at each layer depends polynomially on higher-layer widths, permitting $O_k(1 + p)$ increments and the bound

$$r_k(N)/N \leq O_k(1) \exp(-c_k(\log N)^{1/C_k}).$$

It then reopened the argument for detailed scrutiny, particularly sparse-cube densification, warm-cost detection and the sharpening of the major decomposition needed to avoid feeding excessive preliminary costs into width losses.

The assistant worked out how a positive density surplus could survive the return from sampled paths. Using the reciprocal-sum foundations, it distinguished inexpensive “warm” width losses from more expensive “cold” algebraic costs. Relation removal descended through weighted layers, introduced lower coordinate copies, and preserved higher integer arguments by exact substitution.

A comparison measure separates real and residue copies while retaining inactive parameters. Prime flags and affine-plane counts avoid the full combined modulus. Passive globalization remains ordinary; active replacements use short-variable families, vertical-frequency quotients, and common-site corrections. Representative independence concerns expected payoffs.

Selected affine planes then parameterized residue matching by $x = a + V(B, r)$. Positive averaging fixed the remaining parameters; small boxes restored the determining constraints exactly. The assistant claimed that bounded weight inflation and controlled width losses closed a multiplicative density increment, hence a quasipolynomial iteration. Its subsequent rereading of the foundations retained the conditional shift-comparison input and the additive relative-rank requirement $d + d_0$, rather than treating a merely polynomial rank bound as interchangeable.

The assistant audited the sampling and globalization foundations of the earlier reciprocal-sum argument, with particular attention to repeated width losses. Its proposed accounting charged substantial refinements to strictly lower layers, leaving a rank cut’s own width loss independent of its rank threshold.

The proposed decomposition $D_1 = S(u/(Ha\zeta), (b - A(u))/(v\zeta^j)) + R(u, b)$ separates real interpolation from denominator control. Here S is the slowly varying polynomial written in normalized spatial and residual coordinates, while R is the rational polynomial whose denominators must be controlled. The factors $Ha\zeta$ and $v\zeta^j$ set the normalization scales. Degree induction recovers short directions. Backward lifting requires exact tensor separation because coefficients are unrestricted; constant tensors use pigeonholing.

Failed relations reduce weight- h space and may add weight- $h - 1$ blocks. Exact substitutions propagate upward; shifted coverings preserve surplus. The assistant next checks frequency selection, patch

conversion, cube comparison, convex separation, and independence after conditioning.

VERBATIM EXCERPT

Need check thresholds from *lower or same layer repeatedly* charged as cost of width resetting w_i .
Need independent Q_i . In preparation tests multiple cuts potentially need shrink h width by
direction scaling $1/(||e||)$ etc.

VERBATIM EXCERPT

width budgets additive-by-layer enables $O(p)$ steps polynomial.

38 Auditing density detection and family parameters

The assistant scrutinized the forecasting and globalization machinery inherited from the earlier reciprocal-sum argument. Its analytic inputs included the Leng–Sah–Sawhney inverse theorem [36] and Leng’s efficient equidistribution theorem [37], used in the proposed step-drop framework.

Weighted Cauchy–Schwarz duplicates grid variables and requires sparse boundary control. Modular Fourier decay is truncated by individual character order to avoid one large modulus. Scalar forecasts combine densities, congruences, and exact inactive constraints, with errors fixed before the prime cutoff.

A common top symbol need not give a common marked map after freezing short variables. The assistant constructs a diagonal-identity correction preserving pure-long symbols and aligning dummy projections at one input. Kernel corrections, lattice covers, and family nets follow; pure-long threshold comparisons precede charges for new short lengths.

The assistant audited the proposed bound $r_k(N) \ll_k N \exp(-c_k(\log N)^{c'_k})$ by following weighted determining polynomials through sampling, symbol matching, and ascent. The earlier reciprocal-sum foundations supplied the machinery being adapted. Separate residue moduli for different blocks were intended to prevent lower-layer costs from contaminating higher-layer width bounds. Fixing a common gap from an advance dimension estimate was meant to avoid circularity across density increments.

The key correction distinguished warm approximation from whole-box detection: the detector could be large only relative to normalized chart measure. The proposed argument used a mean-one sparse chart weight, bounded-grid linear-forms estimates, and weighted Cauchy–Schwarz to replace sparse factors by bounded ones before applying inverse theory. The inherited niltest-to-Gowers estimate was specifically attributed to Leng, Sah, and Sawhney, Lemma B.5 [36].

The assistant checks pivot densities, inactive constraints, and cube-image errors in normalized Haar L^1 , charging Fourier complexity to later scales. For passive globalization, degree $d < j$ excludes weight- j arguments, suggesting sharper bounds from separation and modular goodness.

VERBATIM EXCERPT

Cube comparison and dual Gowers bound might require all parameter sides sufficiently large depending on pre complexity, especially very short (t_I) in active or passive paths.

The assistant made the iteration’s resource ordering more explicit, using the absolute increment, symbol calculus, and sampling arguments from the previous bound. It revised the prime-power sublevel argument to use a degree-bounded number of variables, intermediate valuation depths, and a univariate determinant test.

Forward sampling normalizes strict mass by feasible-center proportions. A stop before h divides by $\prod_{i \geq h} e_i$; later discards divide by $\prod_{i > h} e_i$. The assistant checks terminal density and modular flag thresholds through second moments and derivatives.

After preparation cuts, the assistant regenerates samplers from the new root. Dimensions and cut counts precede widths and accuracies. It concludes that $Q_j^{\text{out}} \leq q_j + \text{poly}(p, d_*, q_{>j})$ excludes current width from added losses and permits $O_k(1 + p)$ rounds, while continuing to check circularity and substitutions.

VERBATIM EXCERPT

Need flag bad probability $\leq c_* \mathbb{E}[\phi + \epsilon]$ with centers Haar, not c_* unconditional, due thin strict sets at layers all; cannot pick c_* depending on widths because v_f structural.

39 The proposed quasipolynomial bound

The assistant closed its weighted-cell argument with a sustained audit of the loss hierarchy, using the earlier reciprocal-sum argument. It clarified that canonical-lift identities were required only on the actual orbit and exactly projected output; arbitrary group points instead retained independent site multipliers. A common vertical Fourier expansion preserved the meshes used to group paths.

For transported residual boxes, it used

$$r_h^{\text{old}} = r_h^{\text{new}} + r_{\text{in}} - c - \text{err}$$

and centered the new box near $l_h + c - l_{\text{in}}$, with errors much smaller than the old enlargement gap. It examined possible excessive denominator growth, short-axis effects on symbols, and sparse-chart normalization losses. It also specified that internal residue subdivisions disappear before the active return, and that buffered charts remain comparable in radius with the current width.

The assistant presented the construction as its proposed proof. It asserted that $p \asymp_k 2 + \log(1/\alpha)$, a fixed multiplicative density gain, and only $O_k(1 + p)$ rounds yield

$$r_k(N) \ll_k N \exp(-c_k(\log N)^{c'_k})$$

when $\log N$ exceeds a suitable polynomial in p . It also stated the equivalent quasipolynomial density threshold and the coloring consequence obtained from a largest color class.

VERBATIM EXCERPT

Given incoming center $(l_{\text{in}}) \in [0, 1]^{d_h}$, new width-in τ , want (r_h^{new}) center around $(l_h + c - l_{\text{in}})$.

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